

GUIDE:

Ms. Sai Prasanna

ASSESSING THE EFFECTIVENESS OF GREEN INFRASTRUCTURE ON URBAN FLOODING REDUCTION

Planning Thesis | Urban Flood Management & Sustainability
Study Area: Hyderabad (Ameerpet Ward)



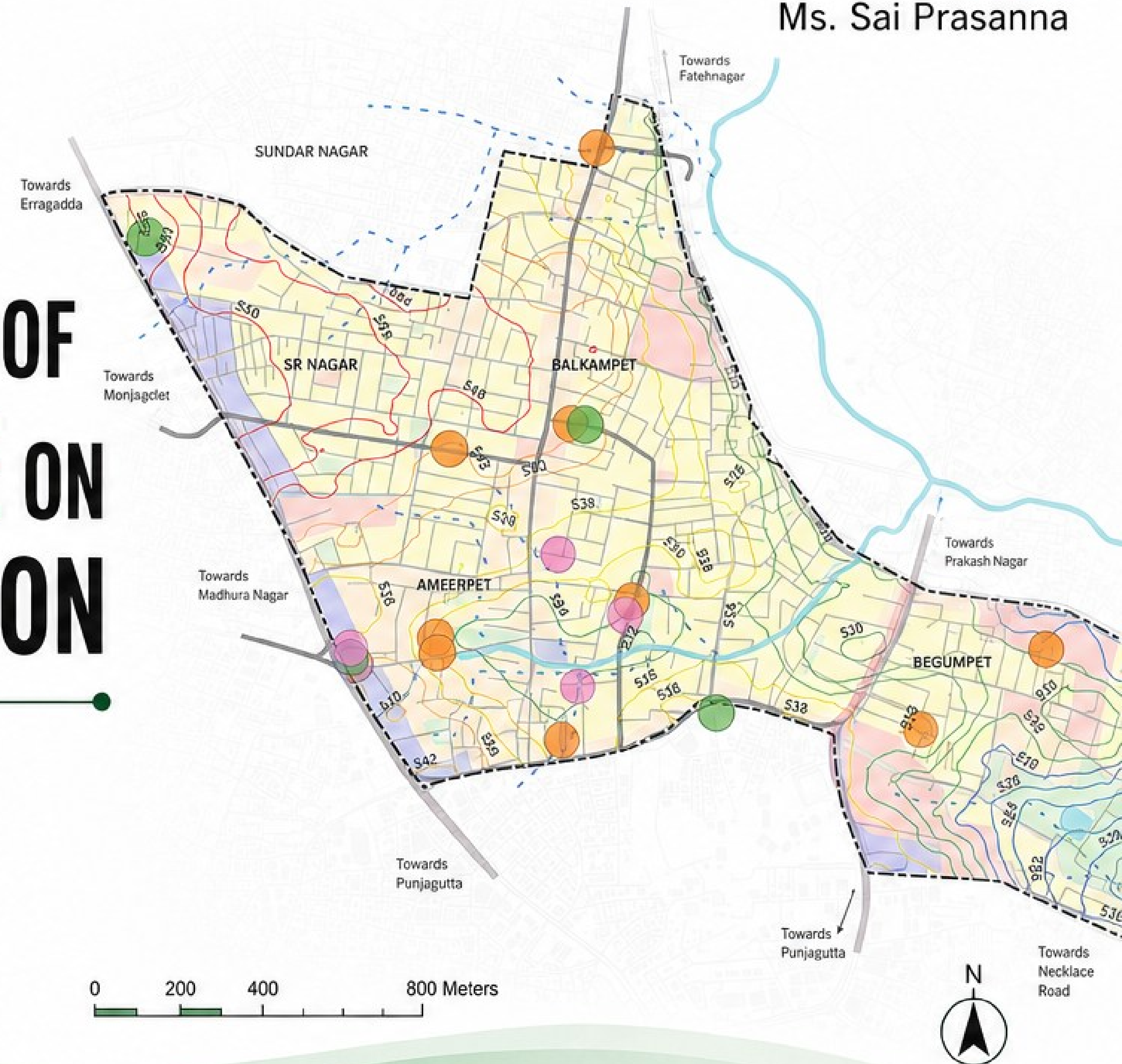
RAINFALL
MANAGEMENT



DRAINAGE
RESILIENCE



SUSTAINABILITY &
GREEN FUTURE



TOPIC:

Assessing the Effectiveness of Green
Infrastructure on Urban Flooding Reduction



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ASSESSING THE EFFECTIVENESS OF GREEN INFRASTRUCTURES ON URBAN FLOODING REDUCTION

Aim

The aim is to minimize and treat storm water in the study area while also providing environmental, social, and economic advantages.

Objective

- To study the concept of green Infrastructure and its applications in world and India
- To study the existing situation of flooding in the study area and issues related to those.
- To implement green Infrastructure techniques in study area

Objective	Data Required	Source	Analysis	Outcome
To study the concept of green Infrastructure and its applications in world and India	• Concept • Importance • Types	Secondary source (Literature Study)	Identifying Parameters	Treat and Manage runoff
To study the existing situation of flooding in the study area and issues related to those.	• Population Density • Rainfall data • Existing Infrastructure • Lanuse and Landcover • Flood prone areas • Topography	Primary Survey Bhuvan TSDPS HMWSSB Secondary data GHMC IMD	Existing scenario	To identify the issues in the study area
To implement green Infrastructure techniques in study area	• Standards • Desktop studies	Primary Survey Secondary data	Giving Short term and long term solutions	Recommendations/ Proposals

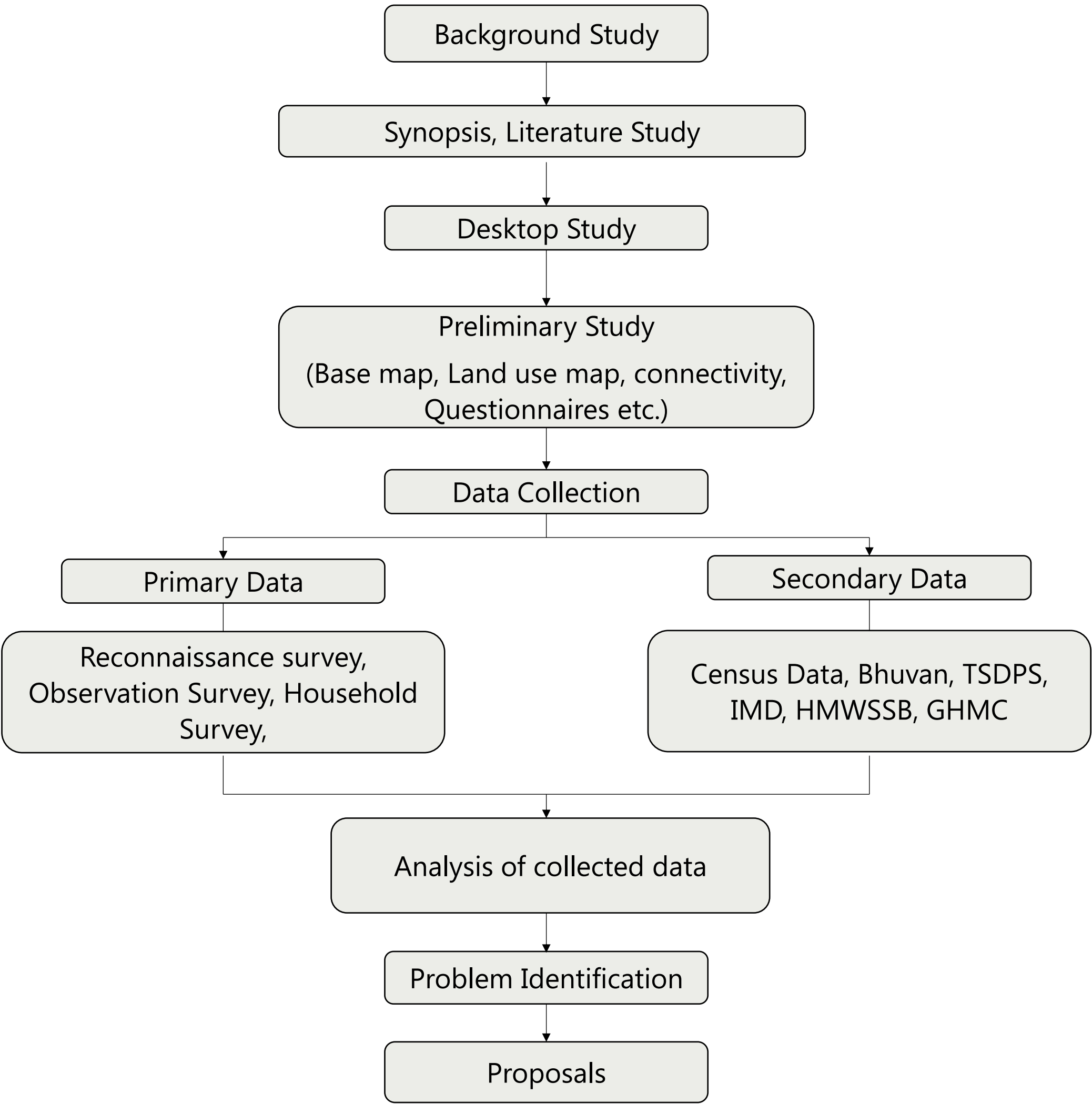
Scope

- Conceptualizing Urban green infrastructure in Indian Context
- Enhancing Urban green infrastructure in future development
- Assessment of green infrastructure in study area

Limitations

In the study only public green infrastructure assets are taken into consideration, private/ individual assets are not taken into consideration.

Methodology



LITERATURE STUDY

GREEN INFRASTRUCTURE

- Green infrastructure is a strategically planned network of natural and semi-natural areas with other environmental features designed and managed to deliver a wide range of ecosystem services such as water purification, air quality, space for recreation and climate mitigation and adaptation.
- Green infrastructure is often contrasted with gray infrastructure, which refers to traditional, built systems like dams, levees, and pipes. While gray infrastructure can be effective at providing certain services, green infrastructure is often seen as a more sustainable and resilient approach to managing the landscape, particularly in the face of climate change.

IMPORTANCE OF GREEN INFRASTRUCTURE

Stormwater Runoff

Runoff, the product of rainstorms or snowstorms, flows over the ground and into drains, sewers, and waterways. The more permeable (or absorbent) the surface, the less runoff there will be. Porous natural landscapes, such as meadows and forests, will readily soak up most of the rain or snowmelt they receive.

Stormwater Pollution

When rainfall hits an impervious surface, it meets whatever pollutants reside on that surface. That could include road salt, sediment, or trash; oil, heavy metals, or toxic chemicals from cars and trucks; pesticides or fertilizers from lawns and gardens; and even viruses or bacteria from animal waste.

Reduced flooding

Caused simply by too much rain on impervious surfaces (not by storm surges or overflowing bodies of water), urban floods can destroy neighborhoods. Green infrastructure reduces flood risks and bolsters the climate resiliency of communities by capturing rain where it falls and keeping it out of sewers and waterways.

Increased water supply

More than half of the rain that falls in urban areas which are covered mostly by impervious surfaces ends up as runoff. Green infrastructure practices reduce runoff by capturing stormwater and allowing it to recharge groundwater supplies or be harvested for purposes like landscaping and toilet flushing.

Health benefits

By reducing air temperatures and pollution, improving water quality, and creating more natural spaces, green infrastructure provides myriad public health benefits. Cooler and cleaner air can reduce heat-related illnesses.

BENEFITS OF GREEN INFRASTRUCTURE

Environmental Impacts

- Green infrastructure installments typically impact the physical environment by decreasing impervious surfaces and creating natural habitat and permeable surfaces.
- Open soil and permeable pavements increase stormwater infiltration and storage capacity; thereby, slowing and reducing stormwater runoff and discharges associated with pollutant loading, flooding, combined sewer overflow (CSO) events, and erosion.

Social Impacts

- Green infrastructure can also have positive impacts on social determinants of health, such as recreation and physical activity, social capital.
- Attractive, accessible greenspace can improve community identity and sense of place, improve aesthetics, and provide a place for gathering and social interaction
- Improved social capital has been linked to reduced stress and improvements in mental health and overall health and well-being.

Economic Impacts

- Green infrastructure costs less than conventional gray infrastructure, provides green jobs and reduces municipal water usage and cooling costs.
- At the household level, this can result in increases in available income for preventative healthcare, healthy foods, and adequate housing, all of which have proven health benefits contribute to overall health and well-being.
- Green infrastructure has also been shown to increase nearby property values, which can be of benefit to businesses and homeowners.

Source: www.epa.gov
environment.ec.europa.eu

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LITERATURE STUDY

TYPES OF GREEN INFRASTRUCTURE

Rain gardens

- Rain gardens—which can be used in a variety of settings, from street medians to small yards—typically feature native shrubs, perennials, and grasses planted in a shallow basin.
- They are designed to trap and absorb rooftop, sidewalk, and street runoff.



Bioswales

- Consisting of long, relatively deep channels of native plants, grasses, flowers, and customized soils that run parallel to parking lots or roads, bioswales can handle large quantities of runoff from large impervious surfaces.



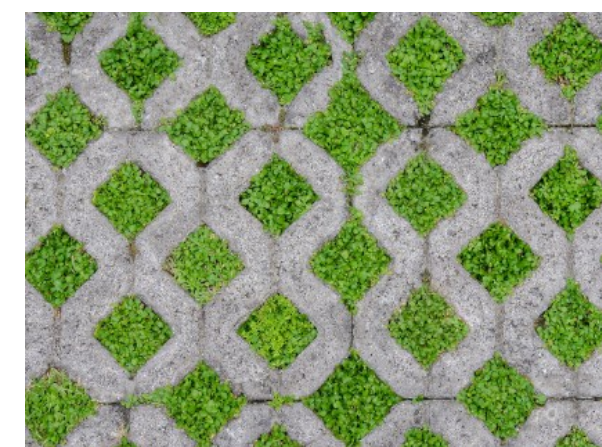
Rainwater harvesting

- Beyond helping to rein in runoff, the capture, storage, and usage of rainfall (a practice known as rainwater harvesting) has the potential to meet 21 percent to 90 percent of a city's annual non-potable water needs, effectively supplying enough water for up to hundreds of thousands of residents.



Green parking lots

- Designed to curb rainfall runoff, green parking lots often feature permeable pavement, vegetated areas around or within their perimeter, and shade trees that can help reduce the urban heat island effect.



Permeable pavement

- A pavement system often used for sidewalks, parking lots, or driveways, permeable pavement (also known as porous pavement) allows rainfall to seep through to underlying layers of pollutant-filtering soil before making its way to groundwater aquifers.

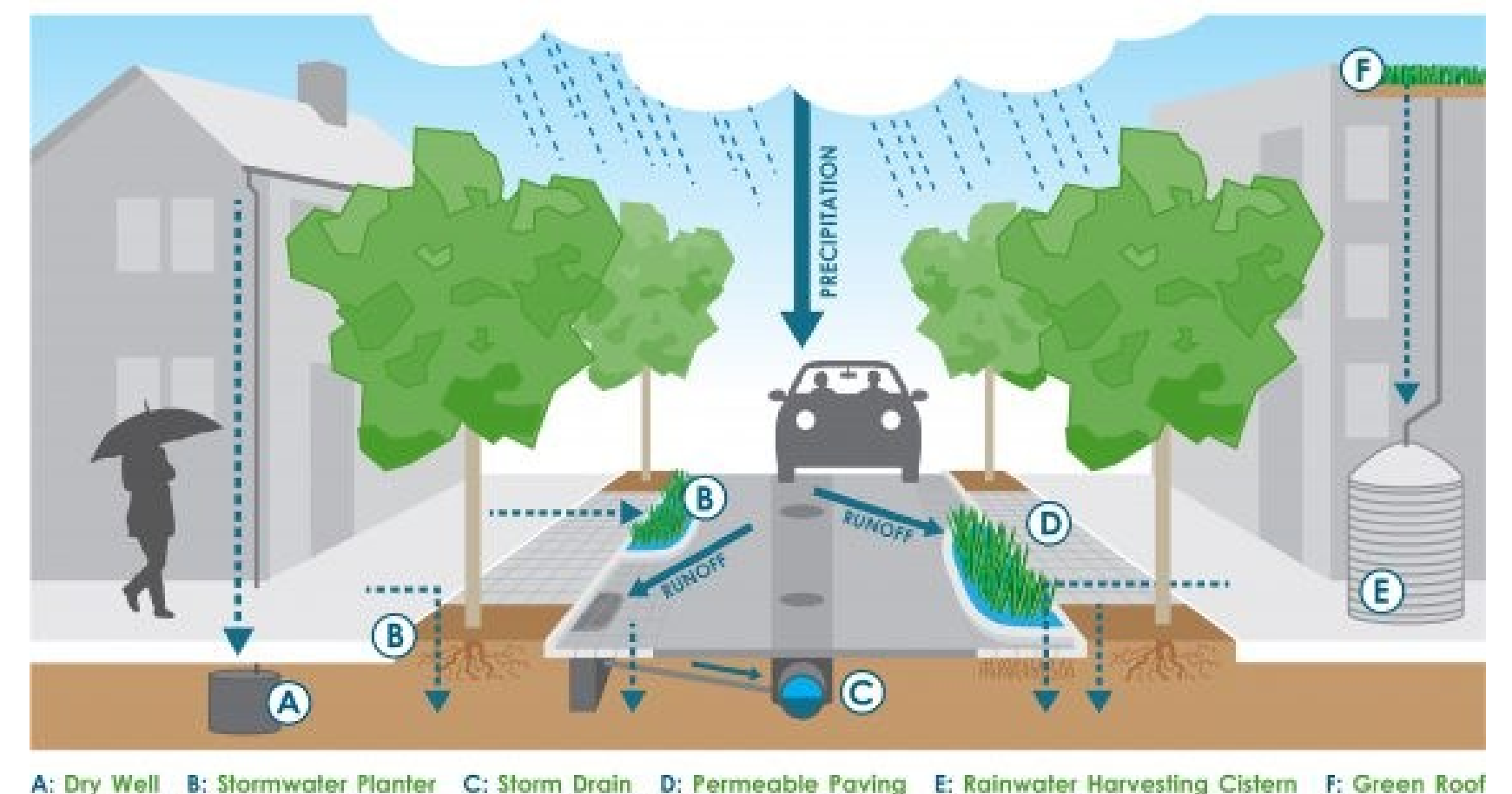
Urban tree canopy

- In addition to cleaning and cooling the air, trees provide a natural stormwater management system. Tree canopies intercept rainfall before it hits the ground.
- By providing surface area branches and leaves for raindrops to land on and evaporate from, a mature deciduous tree can capture as much as 700 gallons of rain a year, while a mature evergreen can absorb up to 4,000 gallons annually.



Green roofs

- A living landscape of vegetation, green roofs provide a verdant oasis for birds, butterflies, and the people who have access to them.
- By providing an extra layer of insulation to a home or building, green roofs slash cooling- and heating-related energy usage and costs.
- They also sequester rain and carbon pollution. Somewhere between 40 and 80 percent of the total volume of rain that falls on green roofs can be retained.



DESKTOP STUDIES

Place	Problem	Objective/ Goals	Solution	Outcome
New York	Sewage overflow	Improve water quality and more sustainable and livable city.	Green streets, Green Parking lots, Green roofs, Rain Gardens, Permeable paving.	Managed the enormous amounts of stormwater volume generated by our nearly 72% impervious landmass.
Atlanta City	Urban Flooding	To alleviate flooding and improve water quality and aquatic habitat.	Bioswales, Green roofs, Rain gardens, Rainwater harvesting, Permeable pavements	These practices treat the stormwater from a total of 1,382 acres of impervious surfaces and remove more than 1 billion gallons of runoff each year from our sewers.
Portland	Urban Flooding	Increase the resiliency of the natural and built environment to increased winter rainfall and associated flooding	Eco roofs, Bioswales, Green Streets	Treating runoff
Katraj, Pune	High Urban Runoff	Reduction in air pollution, To come up with solution and implementation to overcome flood	Green Building Permeable paver roadways	<i>To be implemented</i>
Delhi	Urban Flooding		Bioswales, Permeable paving, Rain Gardens Rainwater Harvesting, Green Spaces/Parks	<i>To be implemented</i>

Parameters	New York	Atlanta City	Portland	Katraj, Pune	Delhi
Rain Gardens	✓	✓	✗	✗	✓
Rainwater Harvesting	✗	✓	✗	✗	✓
Bioswales	✗	✓	✓	✗	✓
Permeable Pavements	✓	✓	✗	✓	✓
Green Streets	✓	✗	✗	✗	✗
Green Parking	✓	✗	✗	✗	✗
Green Roofs	✓	✓	✓	✓	✗
Green Spaces/ Parks	✓	✓	✓	✗	✓

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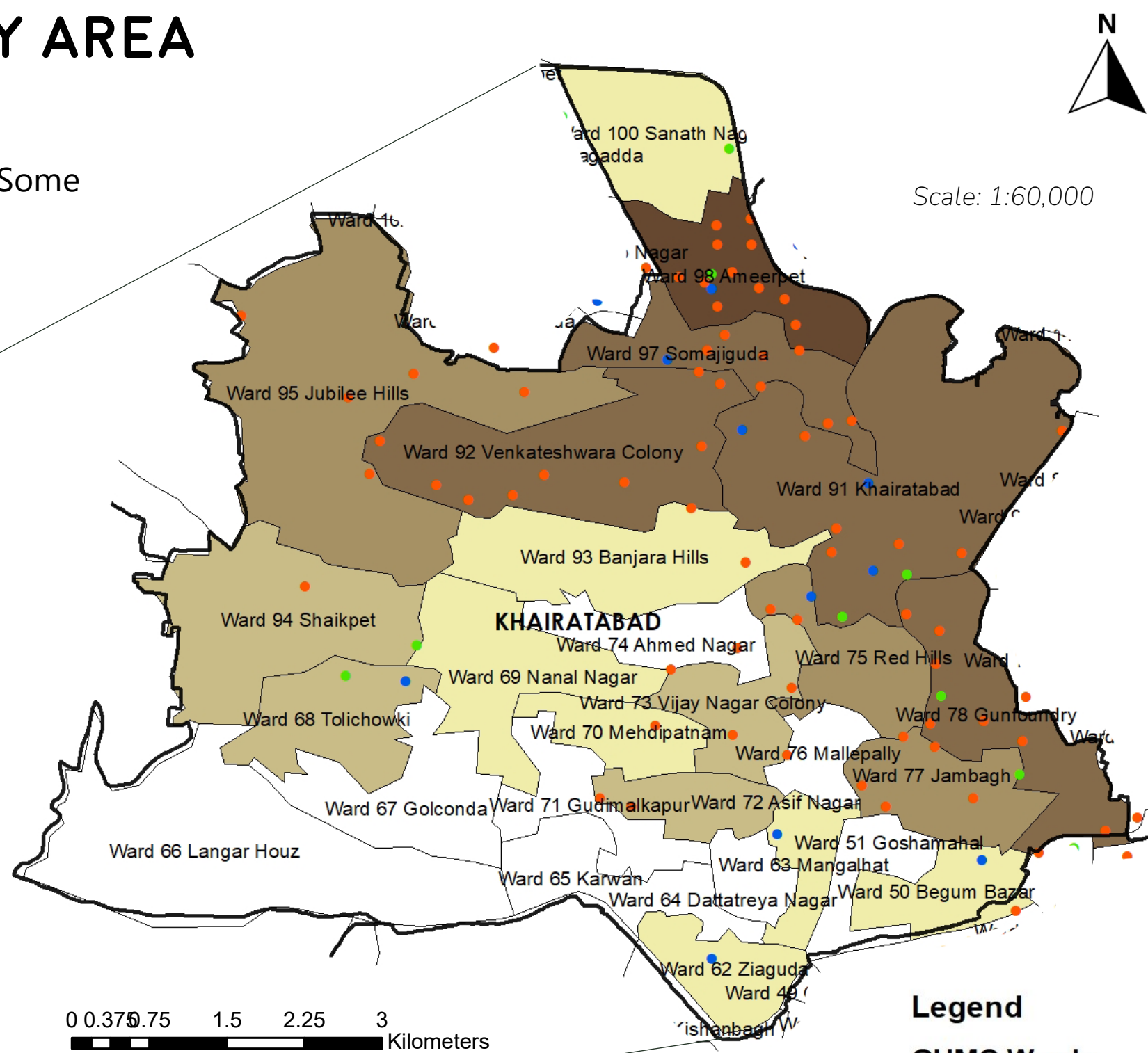
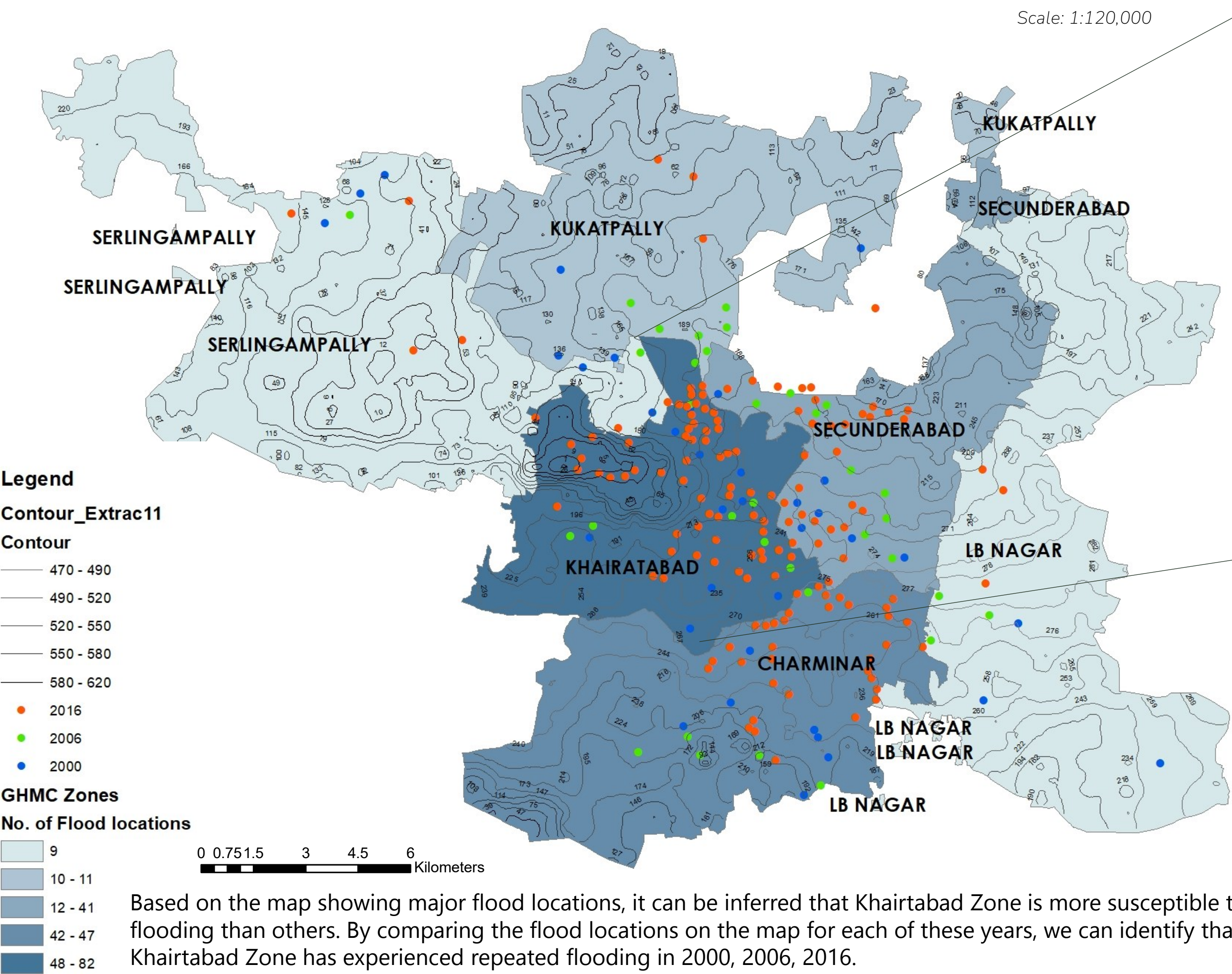
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SELECTION CRITERIA FOR STUDY AREA

Hyderabad, the capital city of the Indian state of Telangana, has experienced several major flood events in recent years. Some of the significant flood events in Hyderabad are September 2021 Floods, October 2020 Floods, August 2000 Floods, September 1908 Floods.



- This map shows the flood locations in the Khairatabad Zone during the years 2000, 2006, and 2016.
- Compared to other wards in the Khairatabad Zone, Ameerpet Ward exhibits a higher concentration of flood locations.
- Ameerpet Ward faced recurring instances of flooding in the years 2000, 2006, and 2016.

(Source: GHMC – 2000, 2006, 2016)

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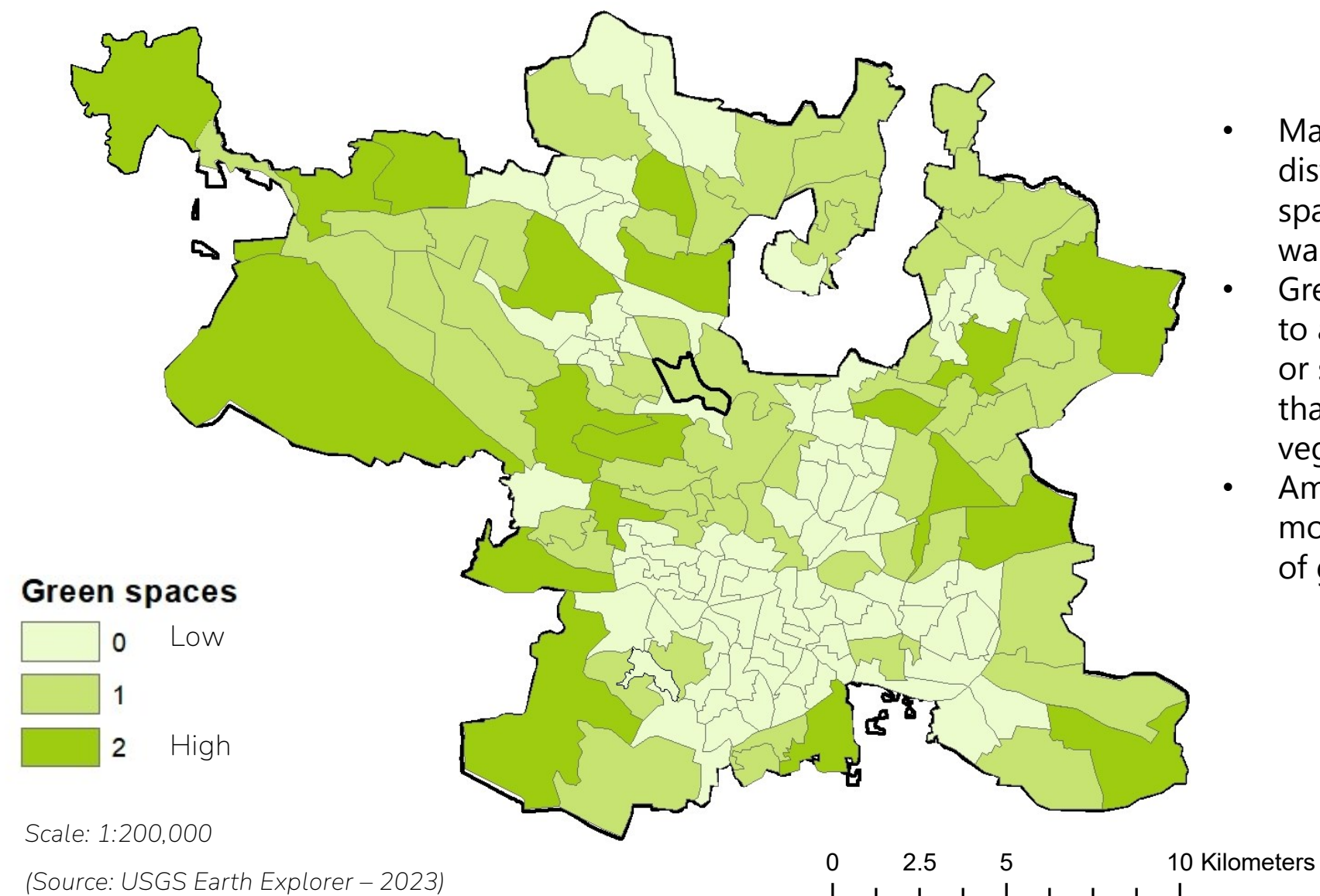
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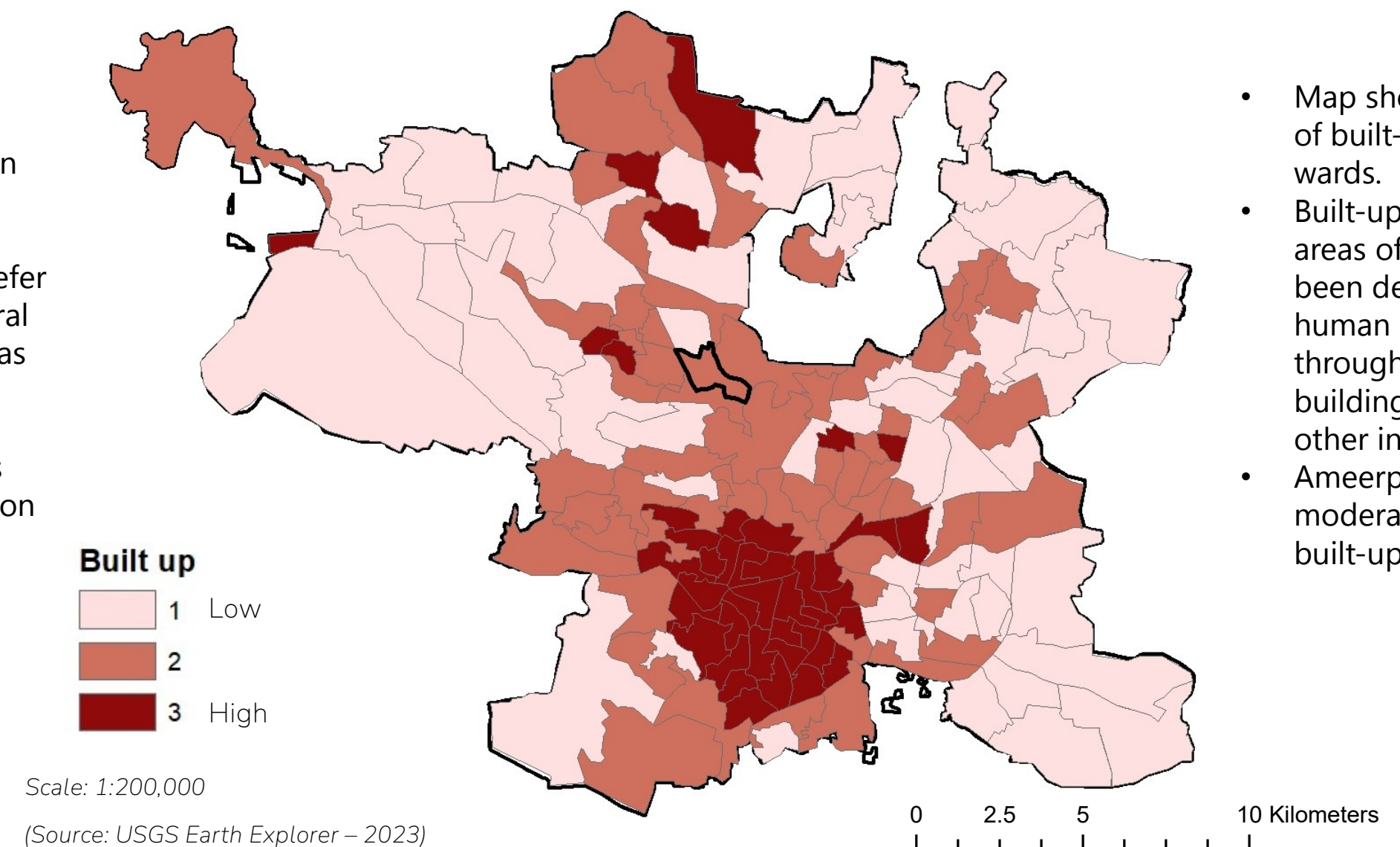


GREEN SPACES



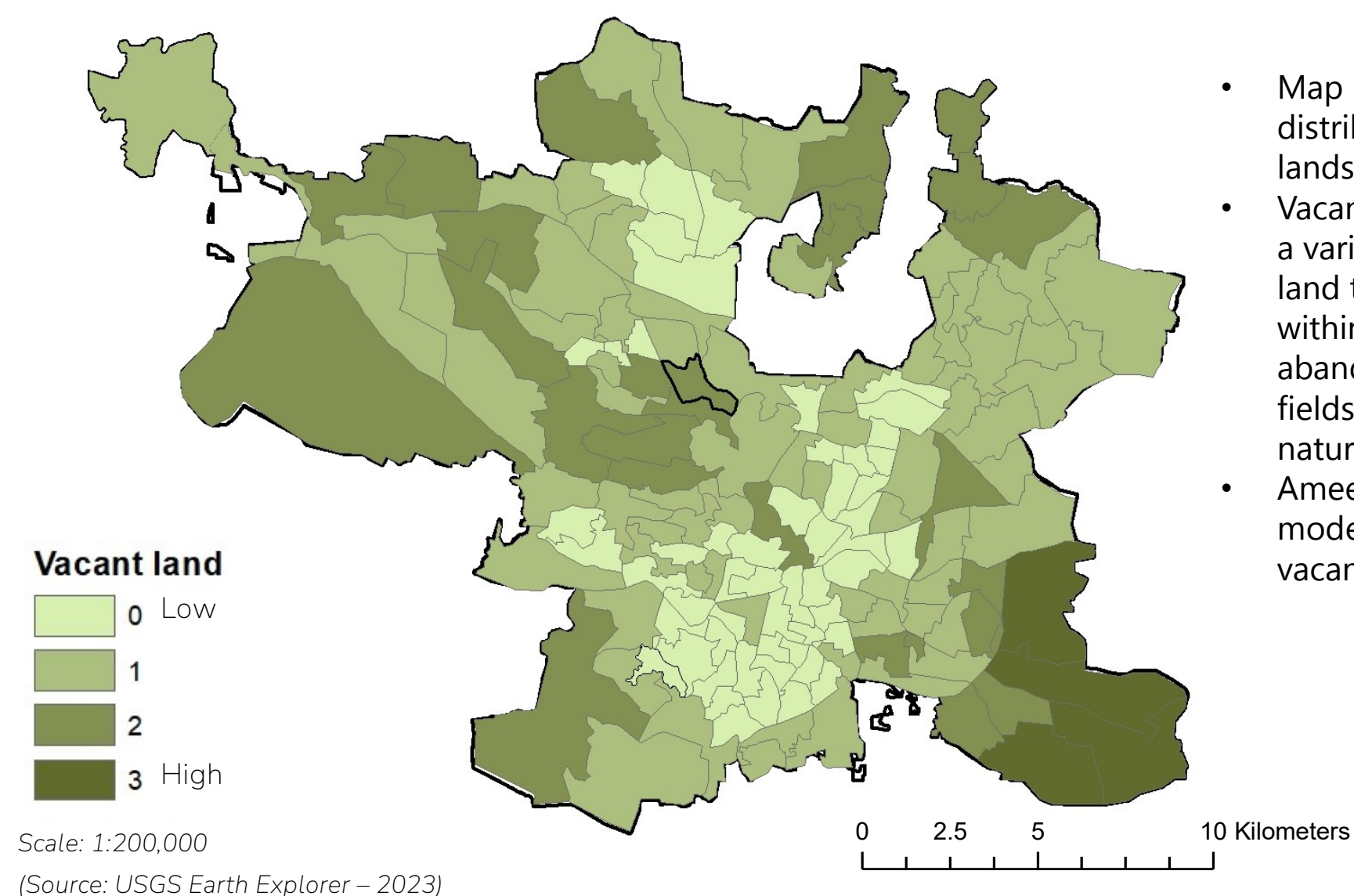
- Map showing distribution of green spaces of GHMC wards.
- Green spaces can refer to a variety of natural or semi-natural areas that are covered in vegetation
- Ameerpet ward has moderate distribution of green spaces.

BUILT-UP



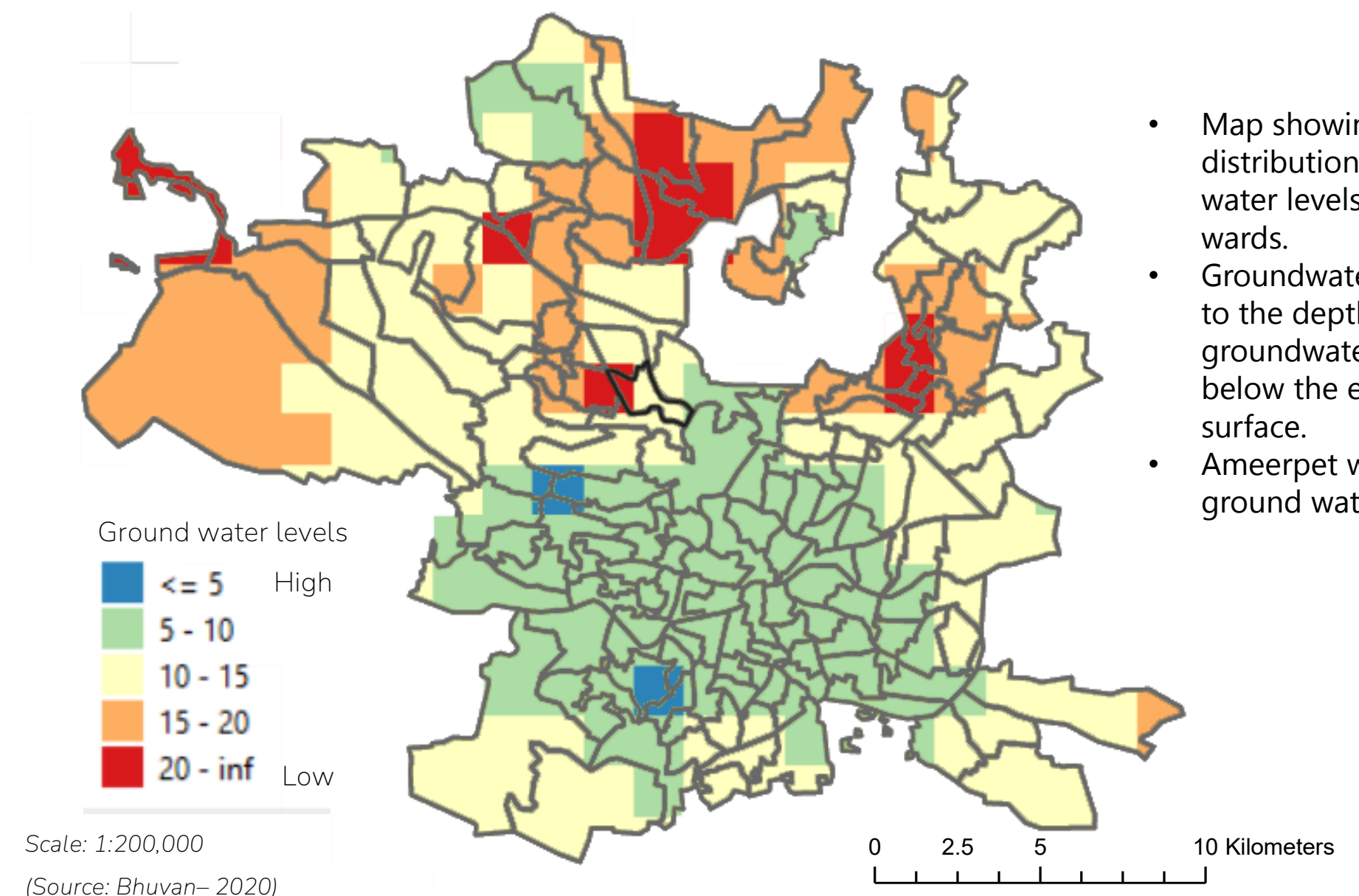
- Map showing distribution of built-up land of GHMC wards.
- Built-up land refers to areas of land that have been developed for human use, typically through construction of buildings, roads, and other infrastructure.
- Ameerpet ward has moderate distribution of built-up land.

VACANT LANDS



- Map showing distribution of vacant lands of GHMC wards.
- Vacant lands can include a variety of different land types, such as lots within urban areas, abandoned agricultural fields, or undeveloped natural areas
- Ameerpet ward has moderate distribution of vacant lands.

GROUND WATER LEVELS



- Map showing distribution of ground water levels of GHMC wards.
- Groundwater levels refer to the depth at which groundwater is found below the earth's surface.
- Ameerpet ward has low ground water levels.

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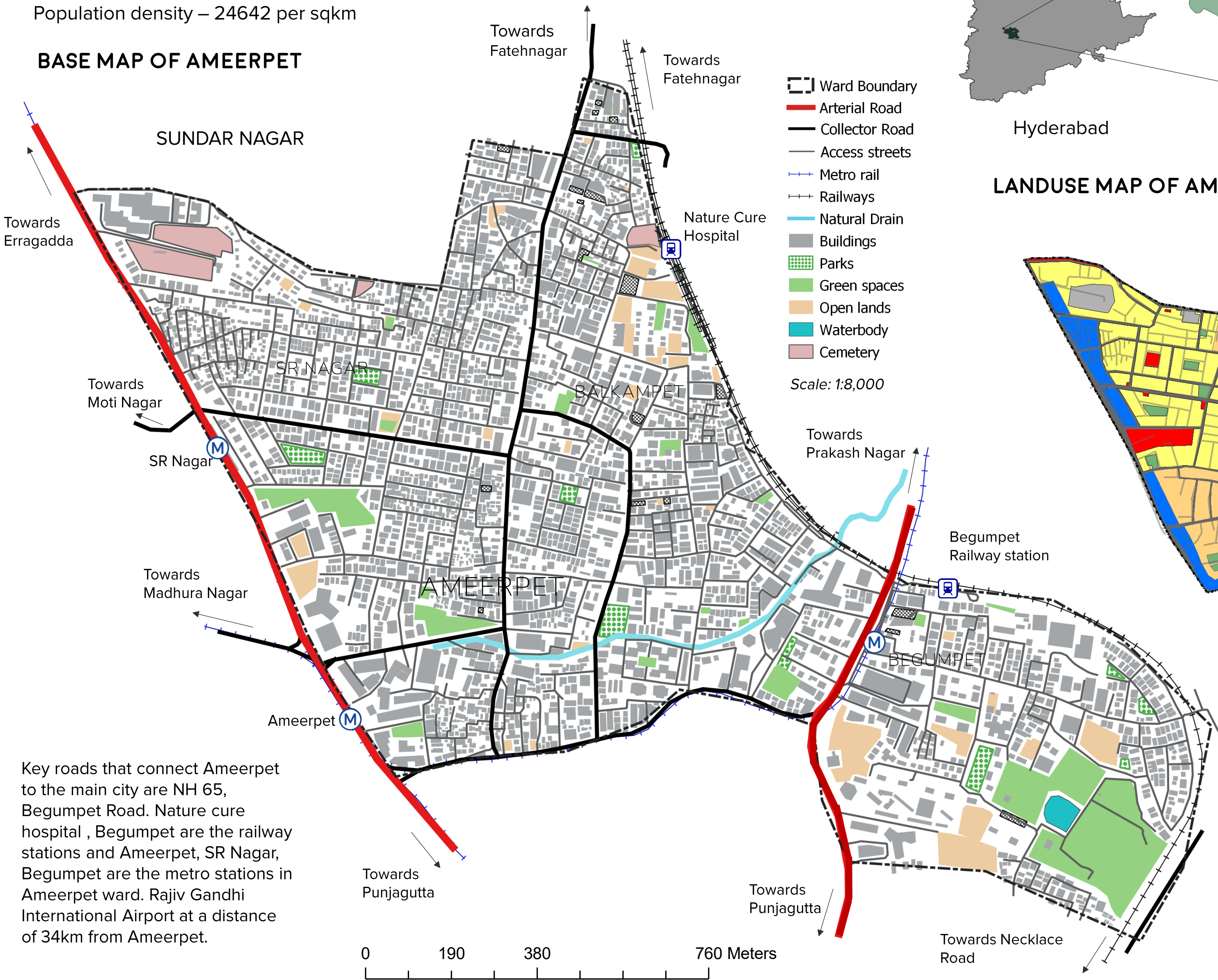
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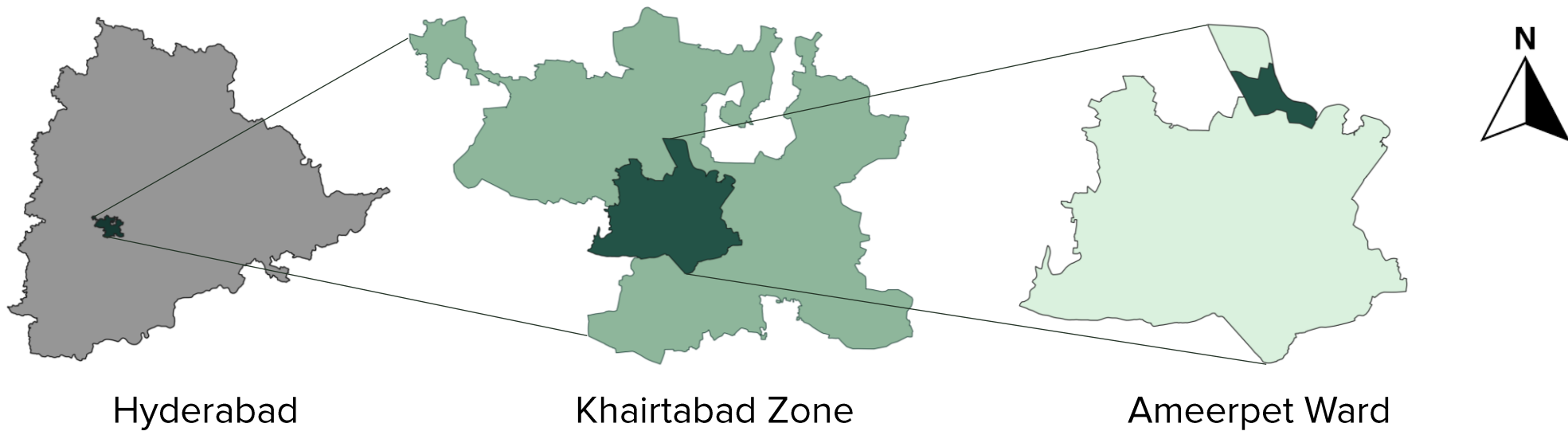
INTRODUCTION TO STUDY AREA

Ameerpet falls in Hyderabad district situated in Telangana state, with the population 72300. The male and female populations are 36445 and 35855 respectively. The area is about 2.4 sqkm
Population density – 24642 per sqkm

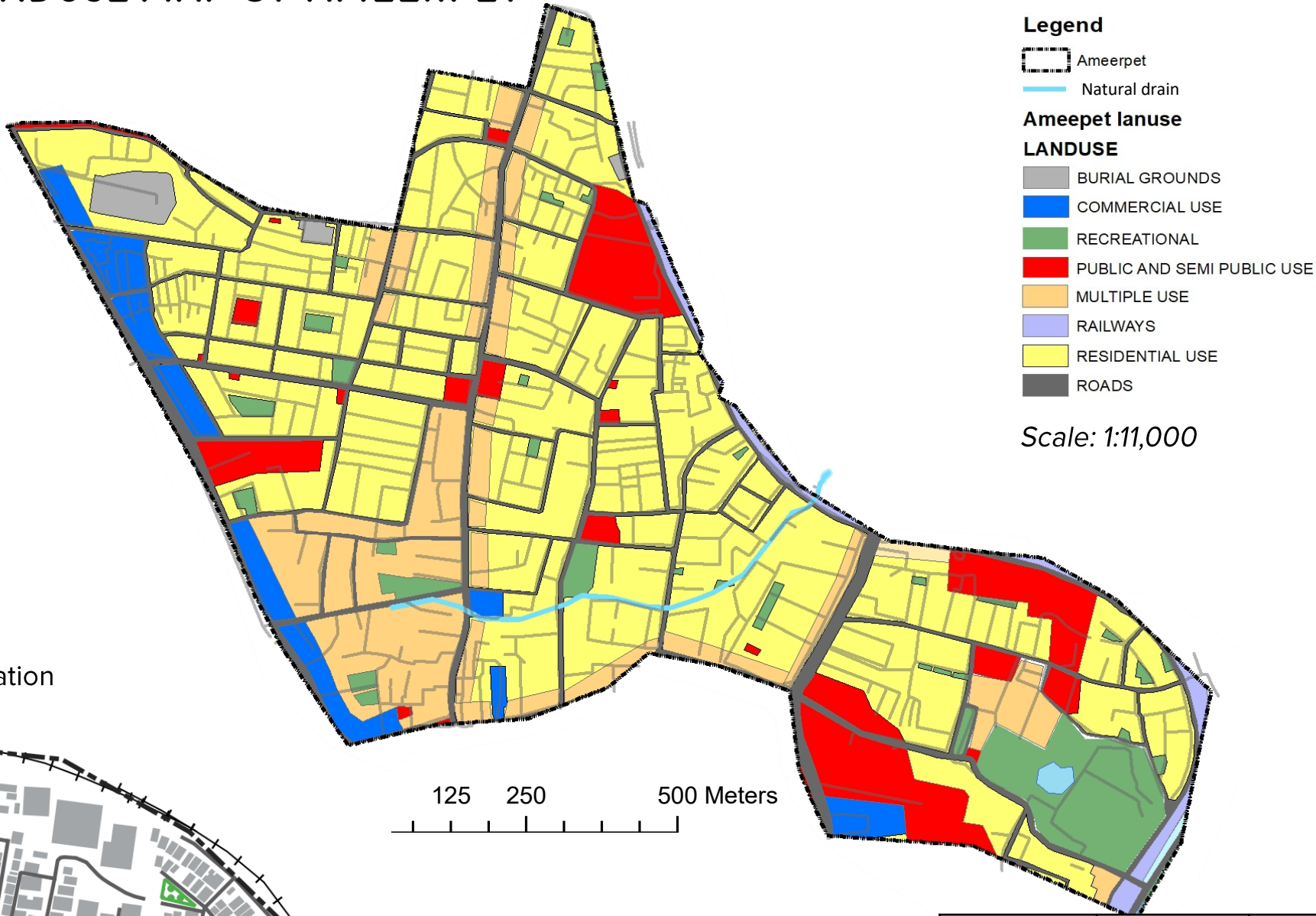
BASE MAP OF AMEERPET



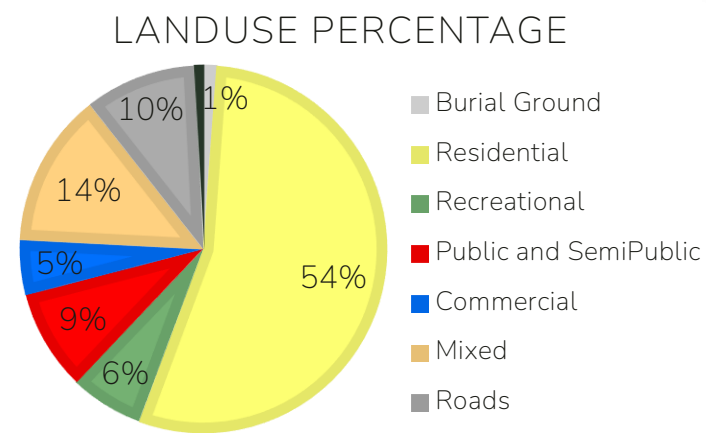
Key roads that connect Ameerpet to the main city are NH 65, Begumpet Road. Nature cure hospital , Begumpet are the railway stations and Ameerpet, SR Nagar, Begumpet are the metro stations in Ameerpet ward. Rajiv Gandhi International Airport at a distance of 34km from Ameerpet.



LANDUSE MAP OF AMEERPET



- Legend**
- Ameerpet
 - Natural drain
 - Ameerpet lanuse**
 - LANDUSE**
 - BURIAL GROUNDS
 - COMMERCIAL USE
 - RECREATIONAL
 - PUBLIC AND SEMI PUBLIC USE
 - MULTIPLE USE
 - RAILWAYS
 - RESIDENTIAL USE
 - ROADS



Major land use type of the study area is residential with the percentage distribution of 54%. Mixed land use percentage is 14% which include commercial, institutional in residential area. Public and semi public land use with the percentage distribution 9% include hospitals, educational institutions.

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ROAD HIERARCHY



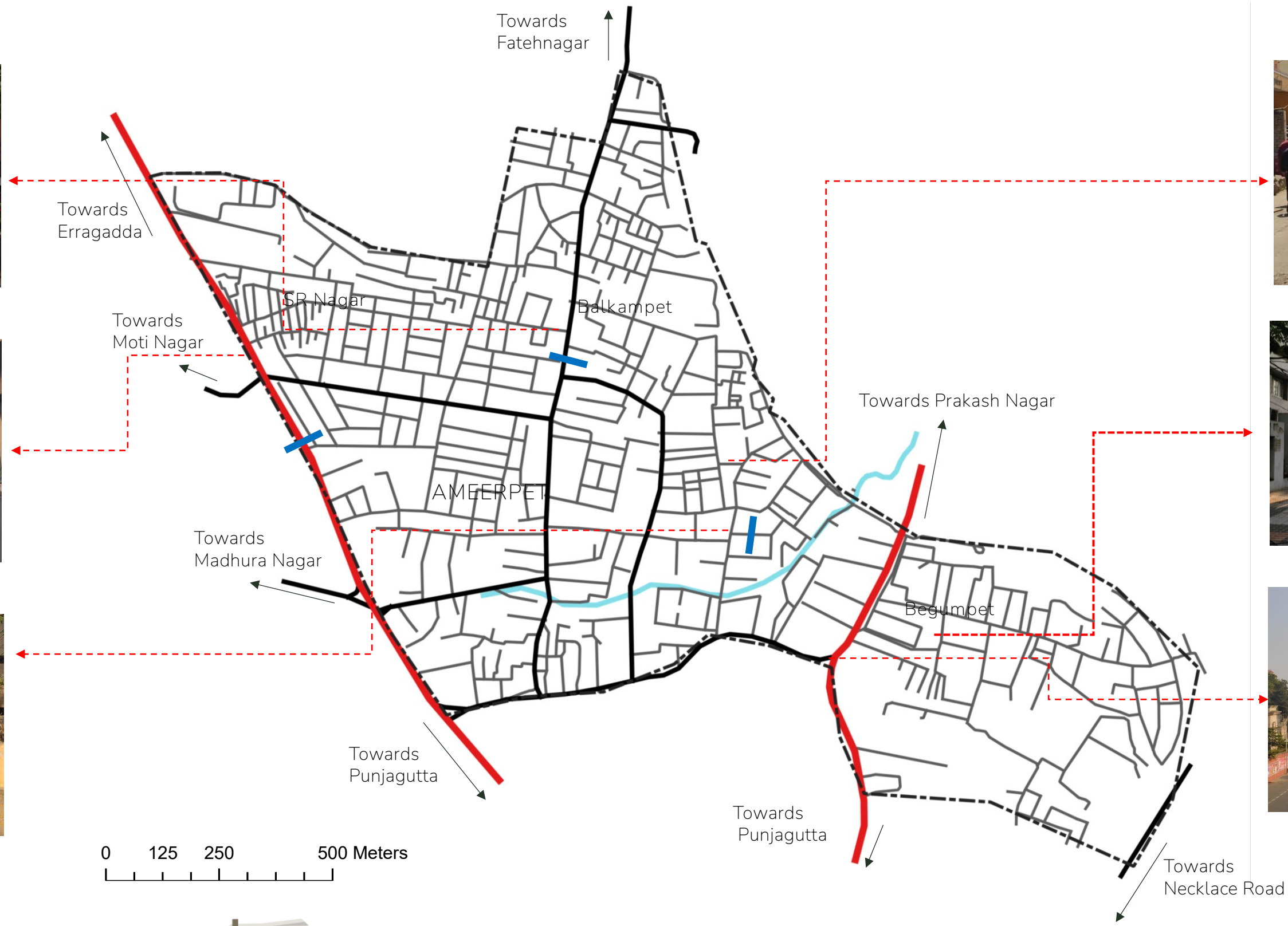
Balkampet Road



NH 65



Local Road



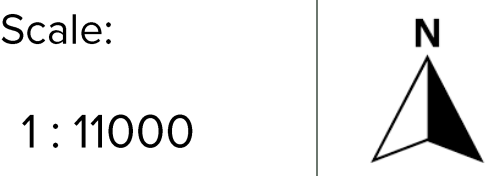
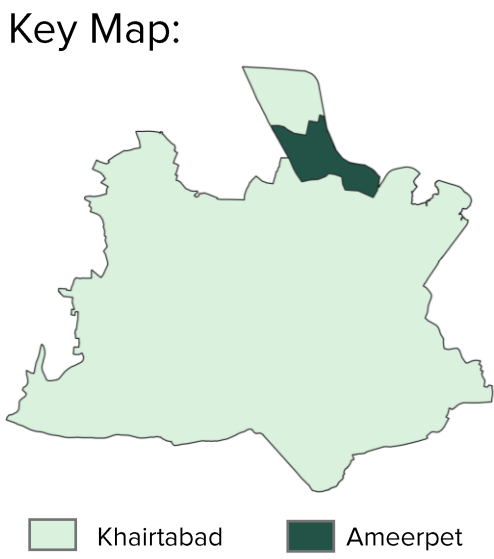
Local road



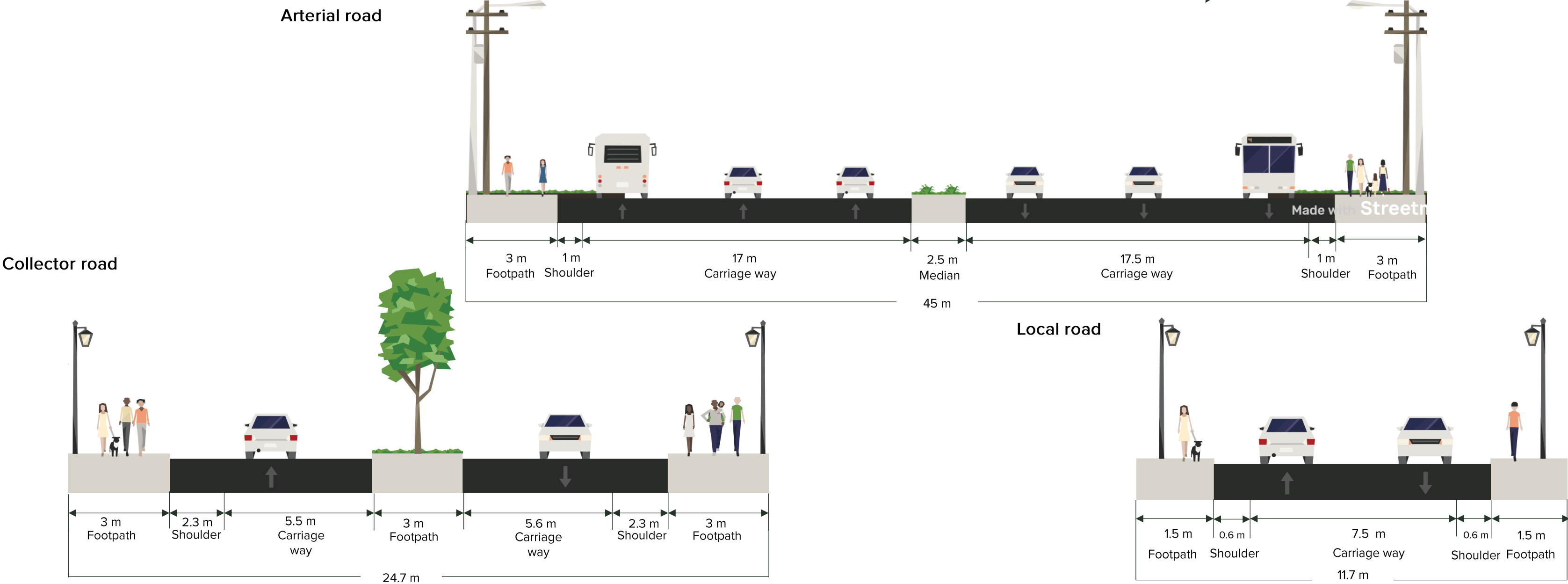
Local road



Inner Ring Road



- Legend:
- Ward Boundary
 - Arterial Road
 - Collector Road
 - Access streets
 - Cross sections



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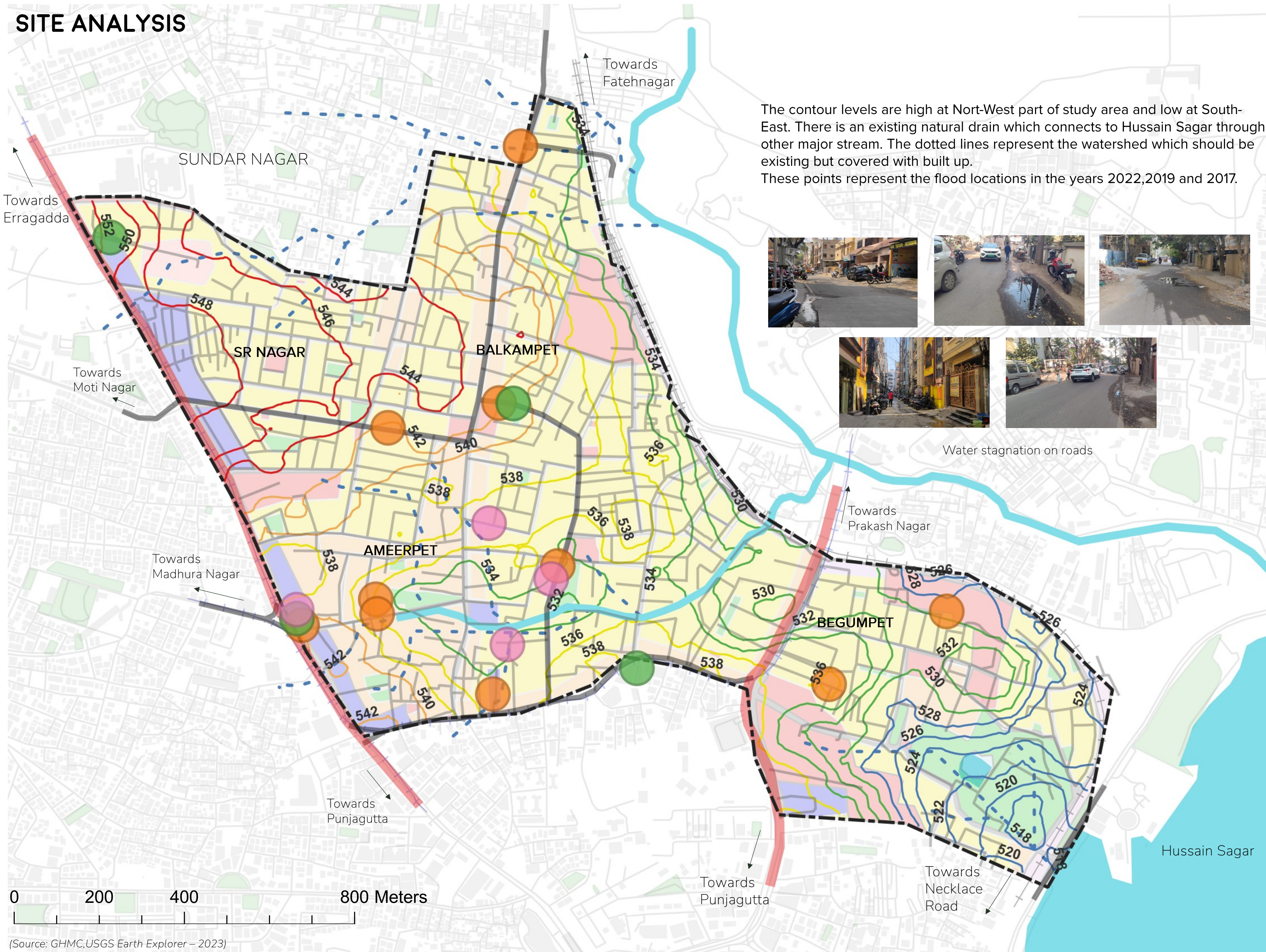
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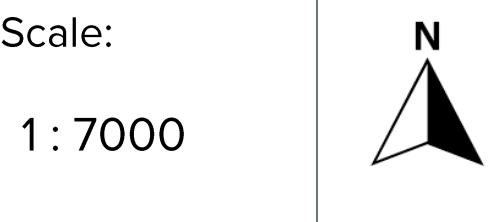
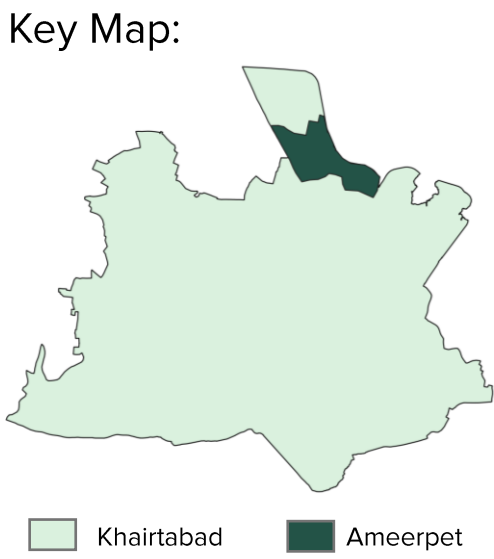
SITE ANALYSIS



The contour levels are high at North-West part of study area and low at South-East. There is an existing natural drain which connects to Hussain Sagar through other major stream. The dotted lines represent the watershed which should be existing but covered with built up. These points represent the flood locations in the years 2022,2019 and 2017.



Water stagnation on roads



- Legend:
- Ward Boundary
 - Contour_Ameerpet
 - 518 - 528
 - 528 - 534
 - 534 - 538
 - 538 - 542
 - 542 - 552
 - Arterial Road
 - Sub arterial Roads
 - Local Roads
 - Metro rail
 - Railways
 - Natural Drain
 - Watershed
 - Flood prone and water logging areas
 - 2022
 - 2019
 - 2017

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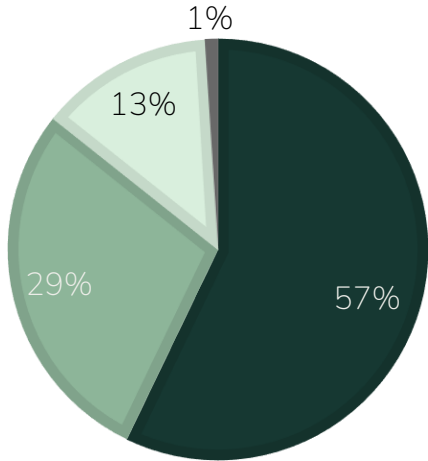
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HOUSEHOLD SURVEY

Property type

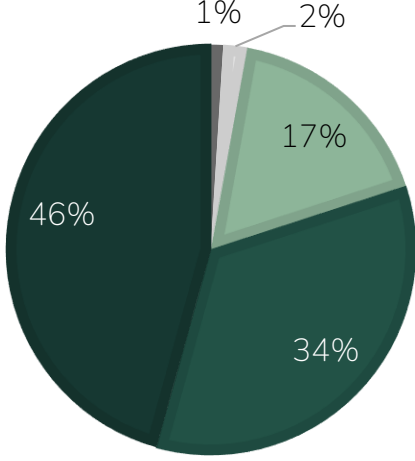
Residential
Commercial
Mixed Use
Public and Semipublic



The establishment of educational and commercial institutions within residential areas has resulted in an increase in mixed land use

Residents Information

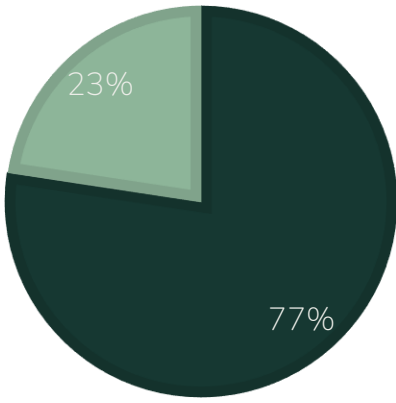
Less than 6 Months
6 months- 1 Year
2- 5 Years
5-10 Years
Over 10 Years



Most of the residents have been in this area over 10 years. The prolonged duration of residency may be attributed to factors such as affordability, convenience, or personal preferences

Flooding

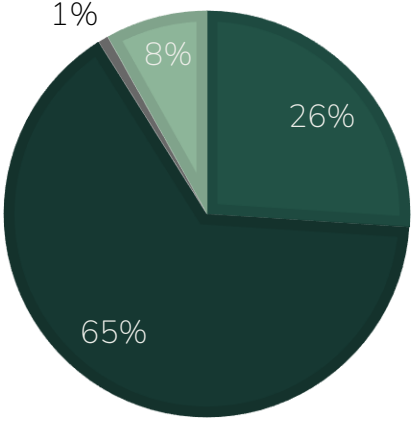
Usually Flooded
Not Flooded



77% of areas are usually flooded and 23% is not usually flooded. This indicates recurring issue of floods within these neighborhoods.

Frequency of flooding

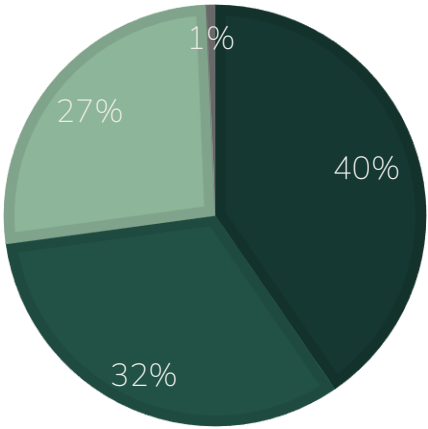
Once a year
Once every 2-5 Years
Several times a year
5+ years ago



Most of the areas gets flooded every 2-5 years and few areas gets flooded once a year

Effects of floods

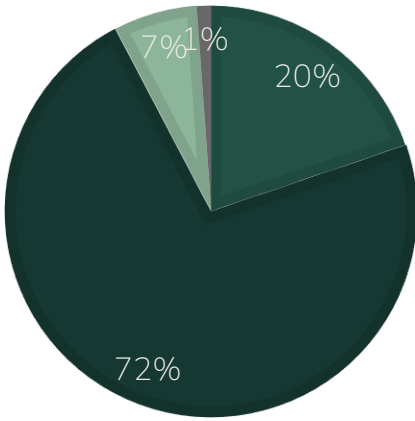
Roads
Footpaths
Buildings/Properties
Open spaces



- Roads and footpaths are flooded mostly
- 27% of have flooding in their properties and buildings

Overflow of water from

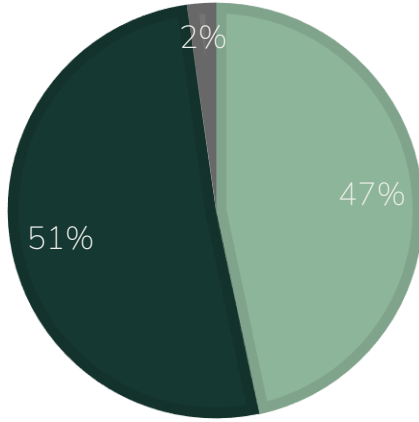
Manholes
Roads
Neighbours Property
Open spaces



- Most of the water that entered their properties come from roads. inadequate drainage infrastructure or improper water flow management on the roads can lead to this

Damage during floods

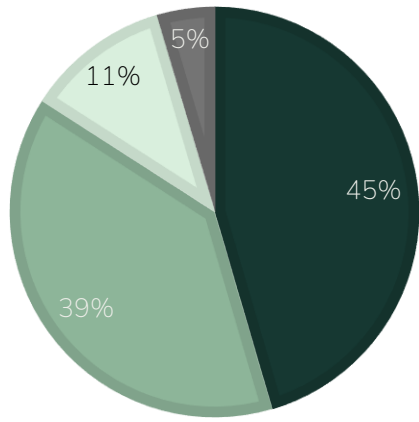
No damage
Minimal damage (less than 10,000)
Moderate damage (10,000 - 50000)



- Areas that have frequent flooding have minimal damage.
- The areas the are flooded rarely no damage

Water scarcity during summer

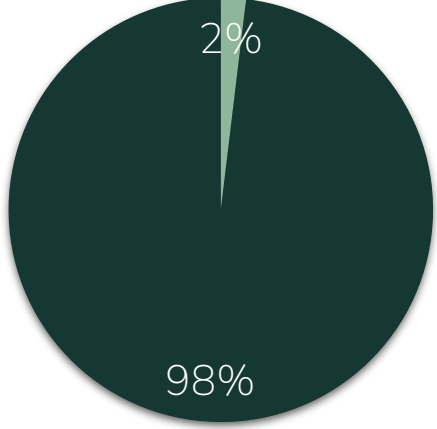
Regularly
Occasionally
Rarely
Never



- 45% of residents regularly face water scarcity issues during summer. 39% of residents occasionally face water scarcity issues.

Rainwater Harvesting

Yes
No



98% of households do not have rainwater harvesting system. This indicates lack of implementation and awareness regarding the potential benefits of rainwater harvesting in the community



Floods at Maitrivanam 2020



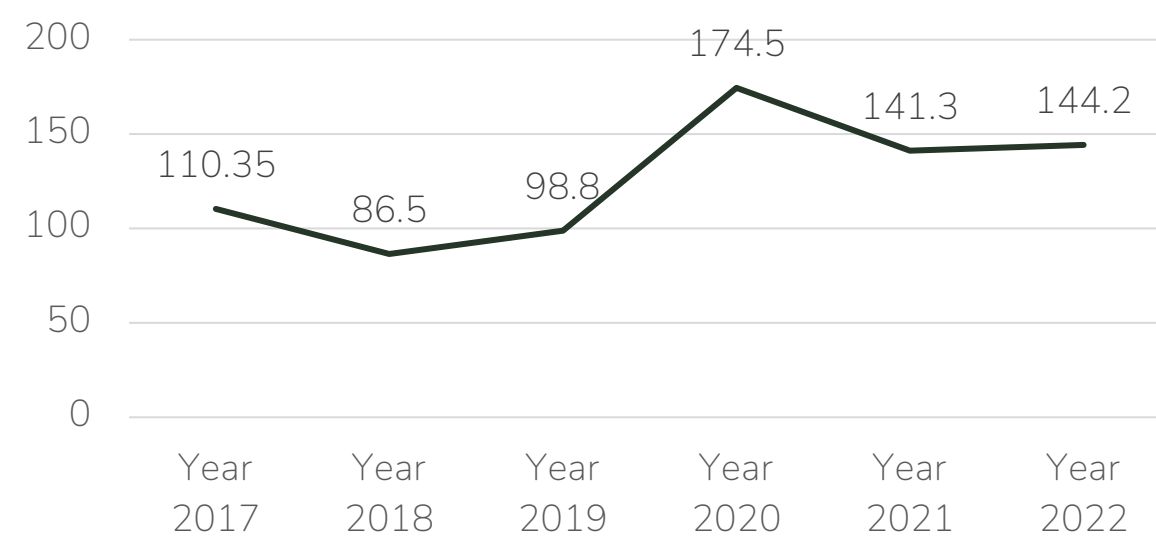
Floods at Maitrivanam 2022

Source: Primary Survey



Rainfall data

Year wise Highest Rainfall in Khairatabad Mandal in mm/hr



(Source: Open data Telangana)

- Graph shows the highest recorded rainfall data for the years 2017, 2018, 2019, 2020, 2021 and 2022 in Khairatabad zone.
- 2020 has highest recoded rainfall i.e. 174.5 mm/hr.

Quantity of Storm Water

The surface run-off resulting after precipitation contributes to the stormwater.

Storm water quantity can be estimated by rational method as below:

Storm water quantity, $Q = C.I.A / 360$

Where,

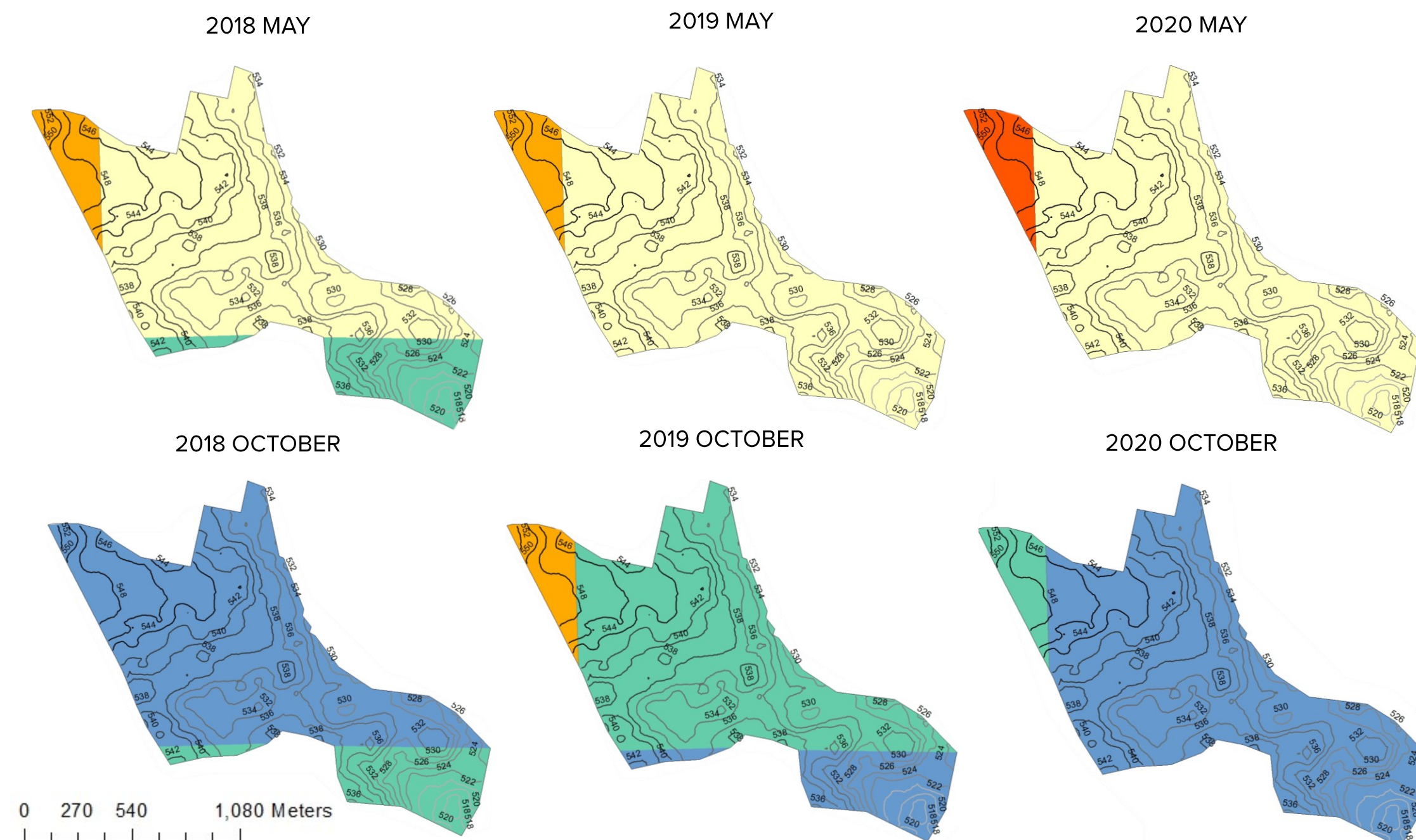
Q = Quantity of storm water, m^3/sec

C = Coefficient of runoff

I = intensity of rainfall (mm/hour) for the duration equal to time of concentration, and

A = Drainage area in hectares

Groundwater level (in m BGL)



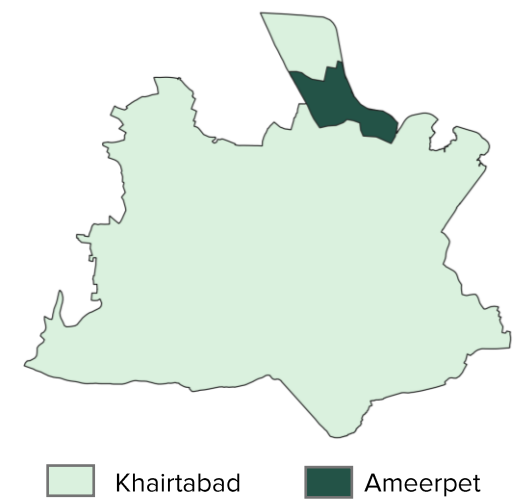
(Source: Bhuvan– 2018, 2019, 2020)

Maps showing groundwater levels of Ameerpet ward for the years 2018, 2019 and 2020 in May and October.

We can observe that depth ground water levels BGL are increasing from 2018 to 2020 in high contour level areas.

Due to increasing in depth od groundwater levels water scarcity during summers occur.

Key Map:



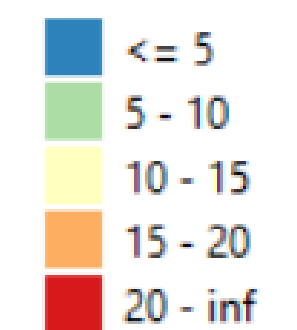
Scale:

1 : 24000

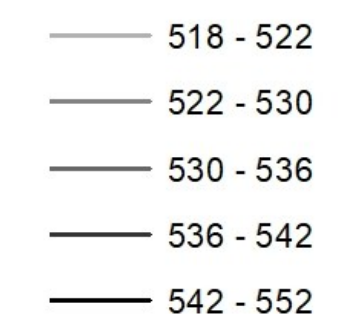


Legend:

Groundwater level (in m BGL)



Contour



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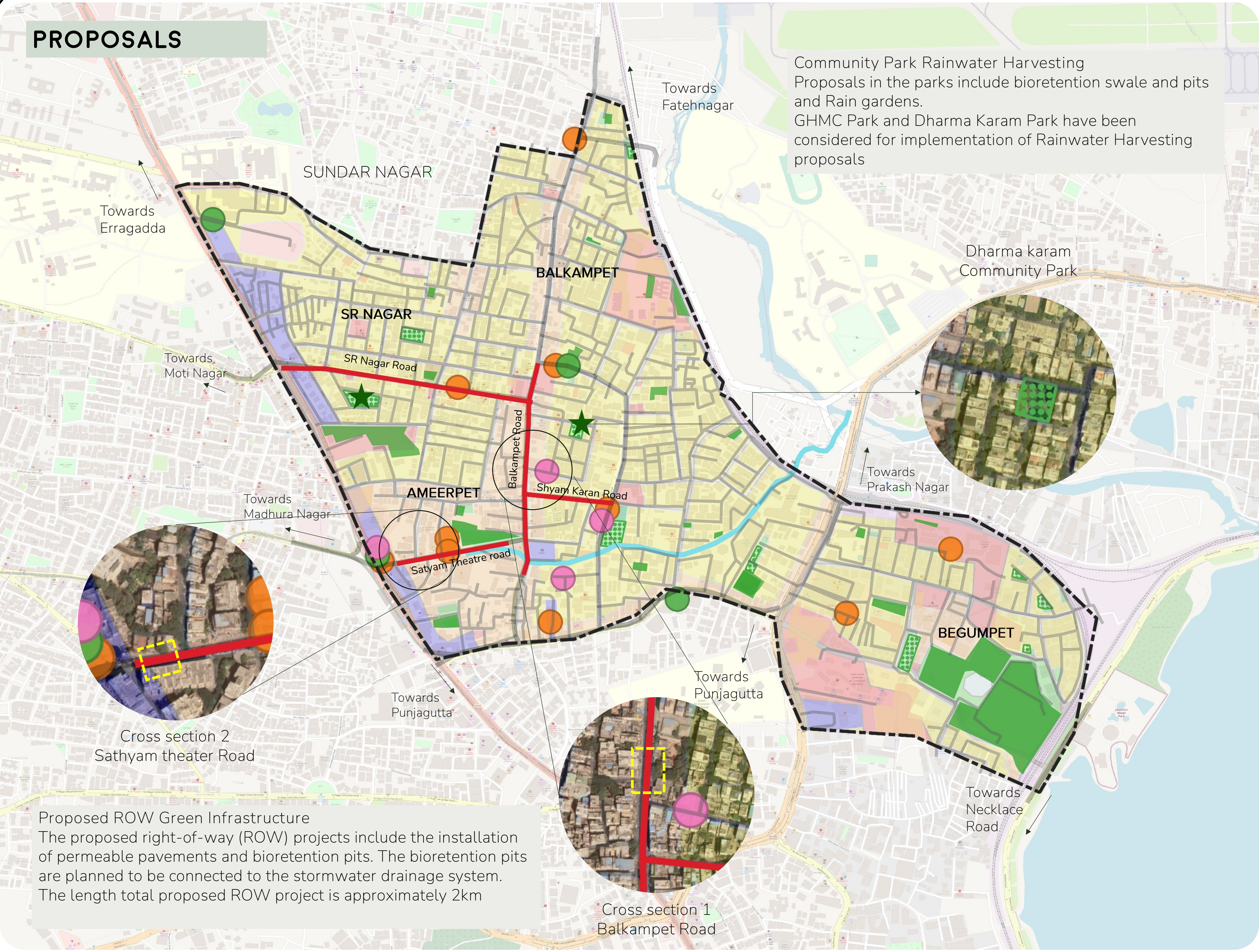
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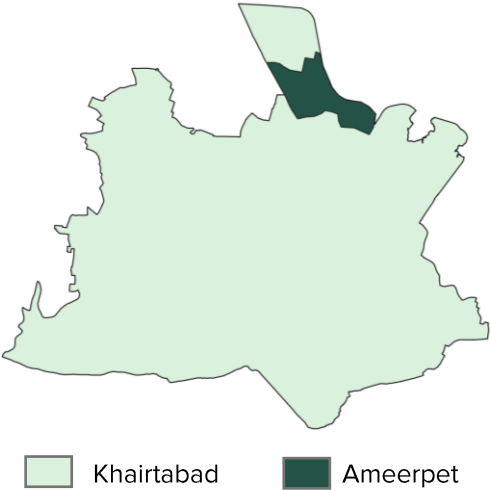
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PROPOSALS



Community Park Rainwater Harvesting
Proposals in the parks include bioretention swale and pits
and Rain gardens.
GHMC Park and Dharma Karam Park have been
considered for implementation of Rainwater Harvesting
proposals

Key Map:



Scale:
1 : 8000



Legend:

- Ameerpet
- Proposed ROW Green Infrastructure
- Natural drain
- Ameerpet lanuse LANDUSE**
 - BURIAL GROUNDS
 - COMMERCIAL USE
 - RECREATIONAL
 - PUBLIC AND SEMI PUBLIC USE
 - MULTIPLE USE
 - RAILWAYS
 - RESIDENTIAL USE
 - ROADS
- Flood prone and water logging areas
 - 2022
 - 2019
 - 2017
 - Community park RWH

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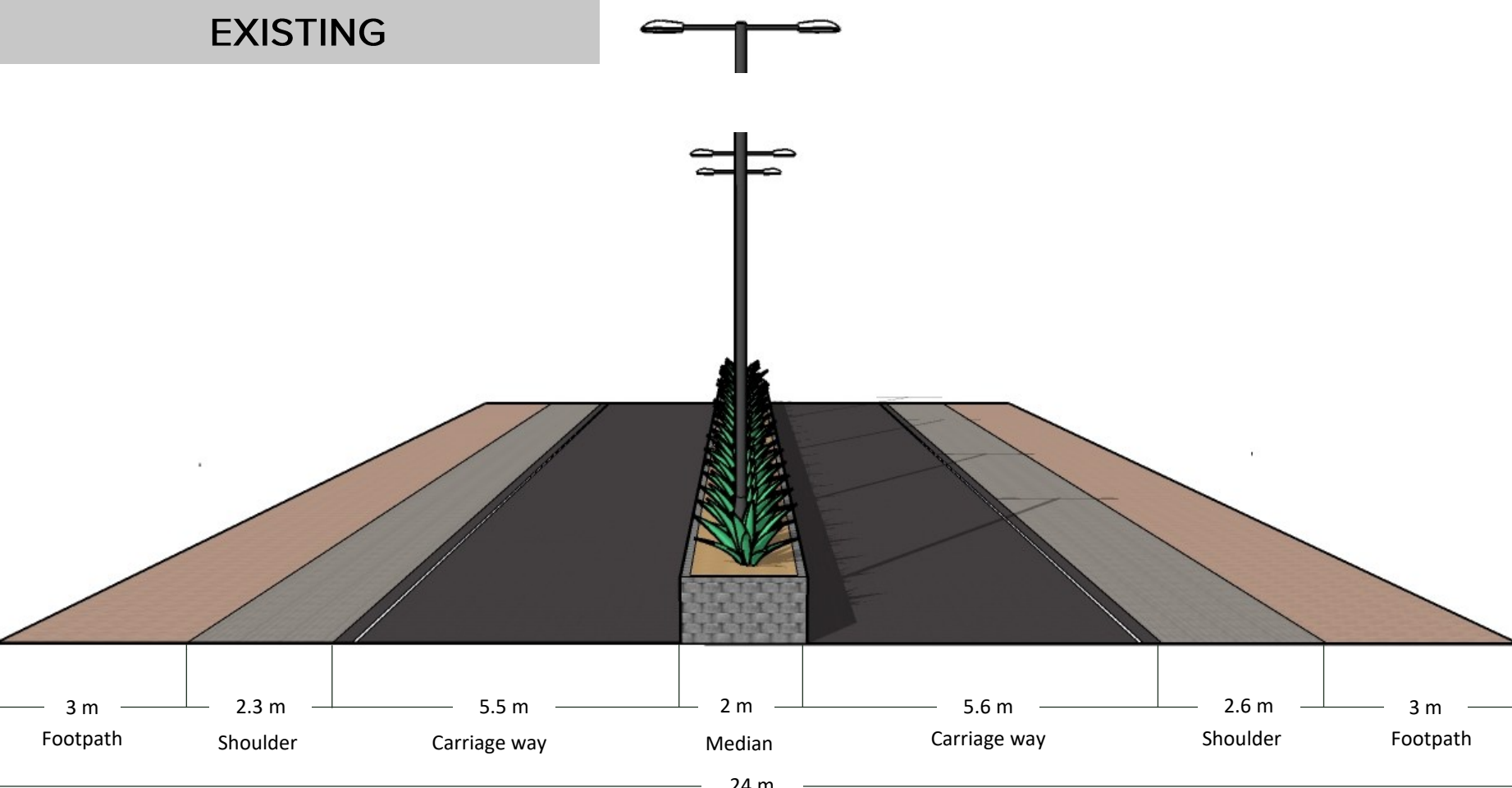
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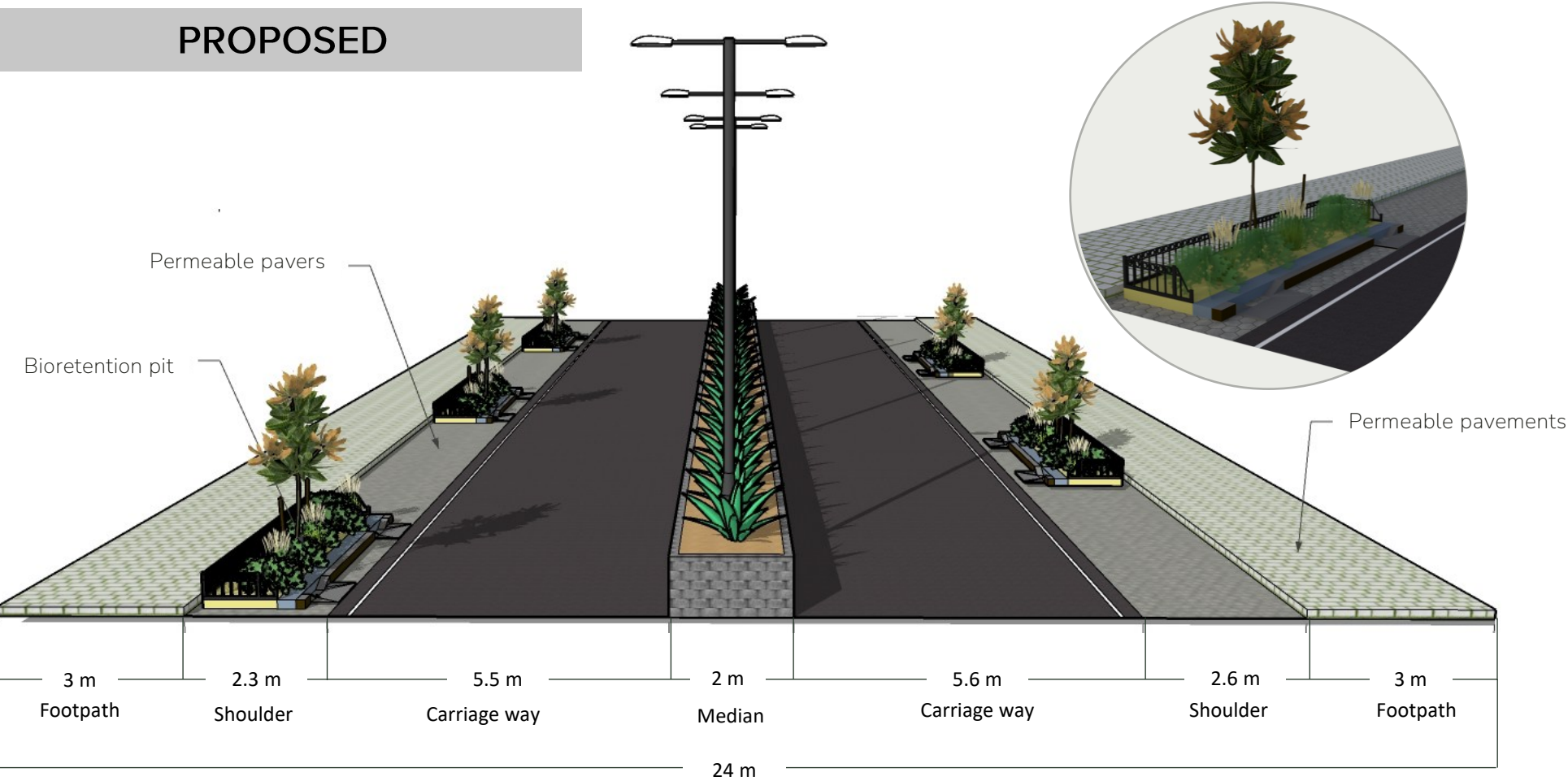


Department Of Urban
and
Regional Planning
Spa, JNAFAU

CROSS SECTION 1 - BALKAMPET ROAD



The Balkampet road currently has a total right-of-way of approximately 24 meters. Each carriage way has a width of 5.5 meters, while the footpath, located at ground level, spans 3 meters and is paved with concrete pavers. The existing shoulder is covered with a concrete base.



The proposed cross section includes permeable pavements and bioretention pits. Bioretention pits have inlets connected to stormwater drainage system. The continuous placement of bioretention pits is proposed for effective stormwater management and infiltration.



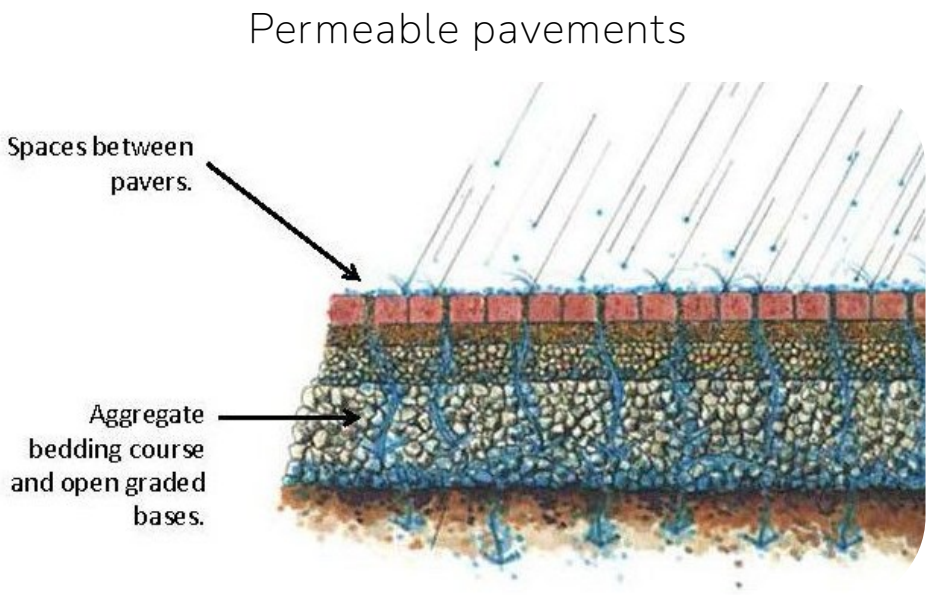
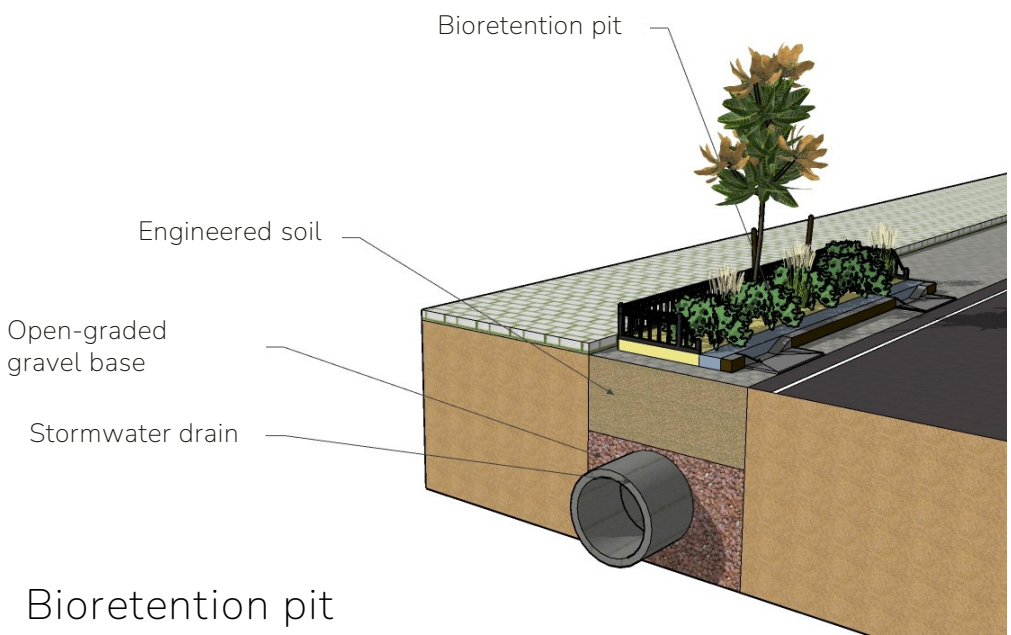
Existing view of Balkampet road



Proposed view of Balkampet road

The total length of proposed permeable pavements and bioretention pits in the ROW is 2km. Balkampet Road, Satyam Theatre road, SR Nagar Road, Shyam Karan Road are the roads with proposed permeable pavements and bioretention pits

- Dimensions of each bioretention pit is around 2 to 4 meters in length, 1 to 2 meters in width, and 0.6 to 1 meter in depth.
- Bioretention pits are often placed within close proximity to each other, typically around 10 to 20 meters apart. This spacing may differ according to the site situation.



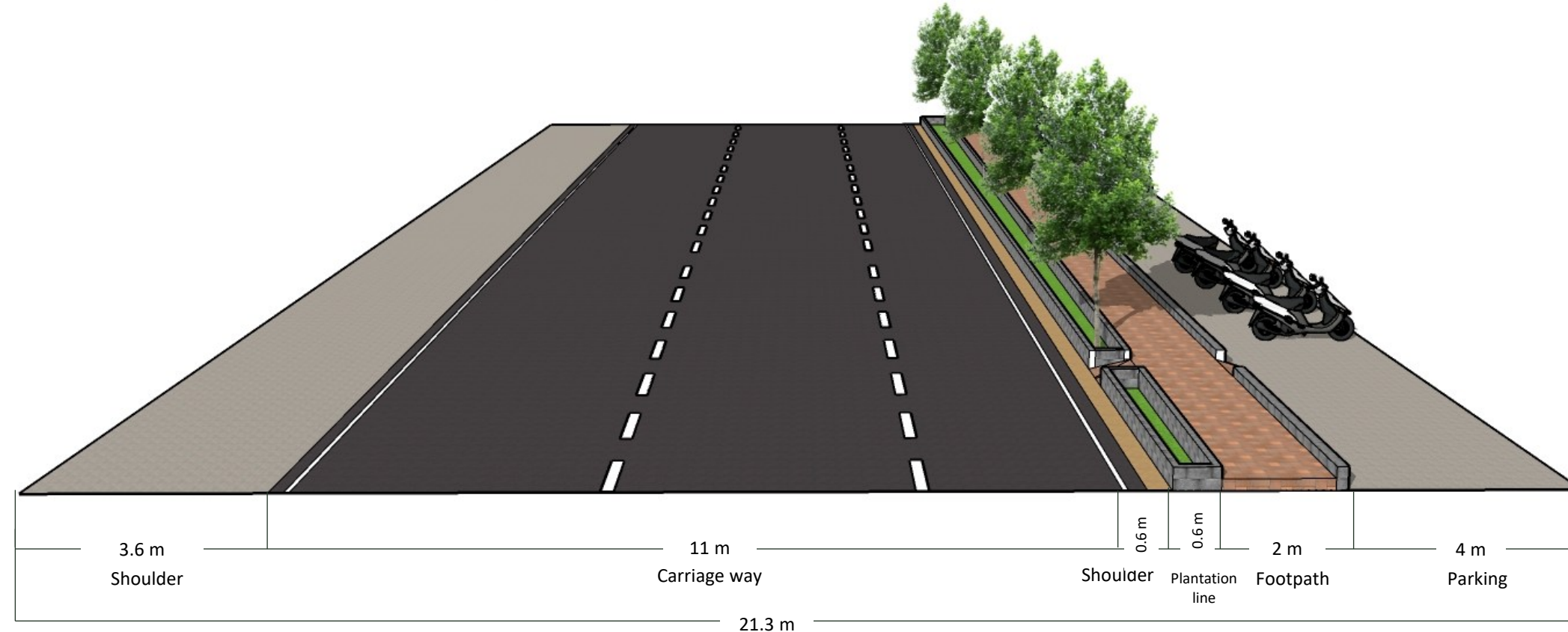
- Balkampet road does not have continuous storm water drainage system. Along the proposed road continuous storm water drainage system is proposed with 600mm to 1000mm diameter. Placed at 1 metre below the surface level incorporated with bioretention pits.



CROSS SECTION 2 - SATHYAM THEATRE ROAD

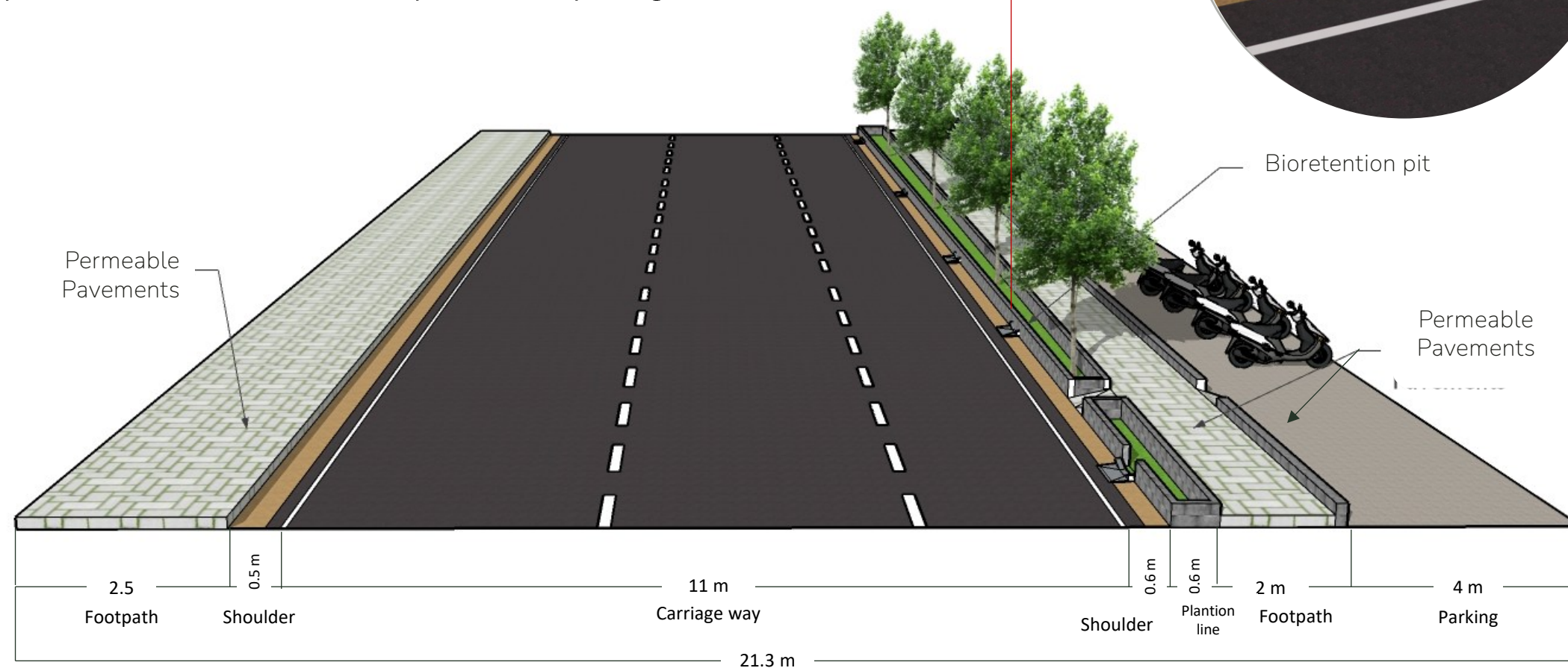
EXISTING

Existing view of Satyam theatre road with total Right of Way 21.3m approximately. Width of carriage way is 11 m. there is no existing footpath left side of the road (towards Satyam theatre) . This road has a plantation line with the width 0.6 m. Footpath and parking space is with concrete based pavers.



PROPOSED

Footpath with the width of 2.5 m is proposed at the left side with permeable pavers . The existing planter line is modified to allow entering stormwater from the roadway. Existing footpath and parking space should be converted to permeable paving.

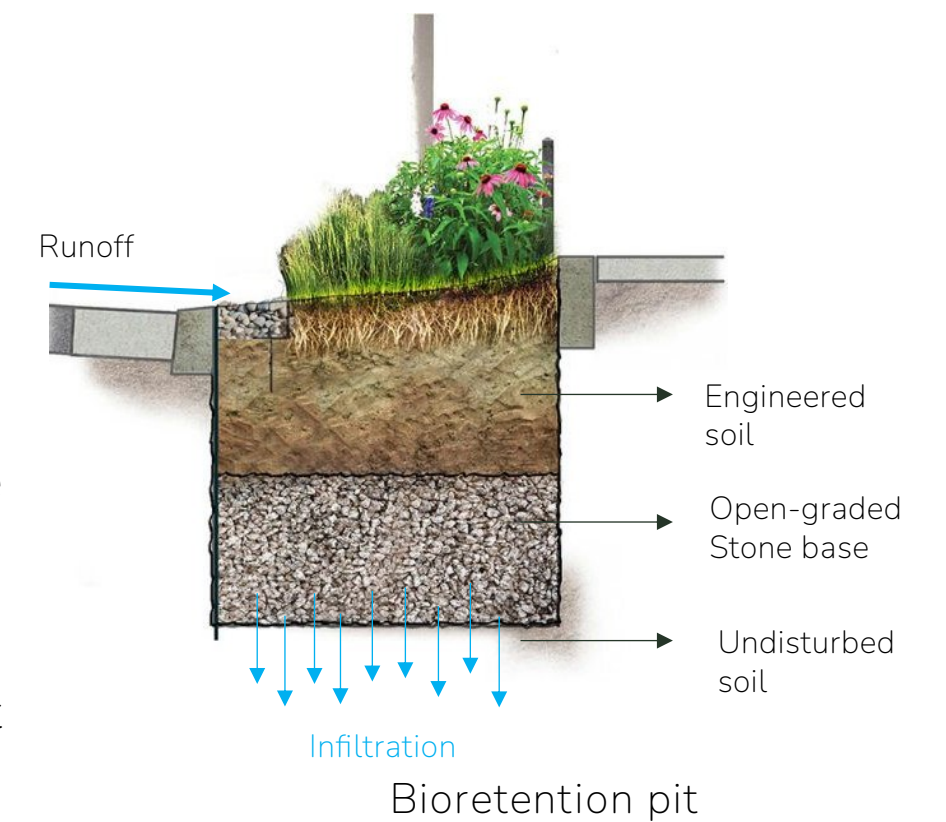


DHARMA KARAM COMMUNITY PARK



Dharma Karam Park Green Infrastructure Proposal

- Total area of the park 1500 sqm.
- Proposed Green Infrastructure in the park include Bioswale and a bioretention pit.
- Bioretention pit is proposed at the center of the park
- Bioswale is proposed along the walking track in the park which can be connected to underground storage tank. Stored water can be later used.
- The width of bioswale is 1-2 m
- For effective stormwater management proposed one other park (GHMC Park) to include green infrastructure such as bioswales, and rain gardens.



Type of Green Infrastructure	Area (approx.)	Per Unit	Total Cost
Permeable pavements	10,000 sqm (107640 sqft)	30/- per sqft	32,29,170/-
Bioretention pit	1000sqm	2390/- per sqft	2,57,23,570 /-
Stormwater drains	2 km	1250 per meter	25,00,000/-
Bioretention swale	125sqm	1055.5 per m	1,31,937/-
Rain Garden	60sqm	200/- sqft	3,29,600/-

TOPIC : ASSESSING THE EFFECTIVENESS OF GREEN INFRASTRUCTURE ON URBAN FLOODING REDUCTION
GUIDE: MS. SAI PRASANNA

K.KUSHI
19011BA010

PLANNING THESIS

DEPARTMENT OF URBAN AND REGIONAL PLANNING
SPA, JNAFAU



RECOMMENDATIONS

RAIN WATER HARVESTING

- The Hyderabad Metropolitan Water Supply and Sewerage Board (HMWSSB) has a rainwater harvesting program in place and encourages the cooperation and participation of all citizens. Government had already made mandatory for construction of Rainwater Harvesting / Conservation Structures in all premises where the plot area is more than 200Sqm to promote water conservation and to protect and to improve the groundwater for sustainable yields in the bore wells.
- Under this program, every household is encouraged to prioritize rainwater harvesting within their premises. By collecting and utilizing rainwater, households can make a significant contribution to water conservation efforts and protect the availability of groundwater resources and mitigate urban flooding
- Environmental Benefits: Rainwater harvesting helps reduce stormwater runoff, which can carry pollutants and contribute to urban flooding. By capturing rainwater, we can minimize the strain on stormwater drainage systems and protect our local water bodies from pollution.
- Economic Benefits : Implementing rainwater harvesting can lead to significant cost savings in the long run. By reducing our dependence on external water sources, we can lower our water bills and decrease the burden on municipal water supply infrastructure.
- Social Benefits: By promoting the importance of rainwater harvesting and its benefits, communities become more knowledgeable about water resources, conservation, and sustainable living practices. This awareness can lead to a broader understanding of the interconnectivity between water, ecosystems, and human well-being.

GREEN ROOFS

- The implementation of green roofs is necessary due to their numerous benefits, making them a crucial solution for addressing urban flooding and promoting a more sustainable and resilient environment.
- Green roofs absorb and retain rainwater, reducing the volume of stormwater runoff. This helps alleviate the burden on stormwater drainage systems and reduces the risk of urban flooding during heavy rainfall events..
- By absorbing and storing rainwater, green roofs help mitigate the impact of urban flooding. They act as a natural buffer, slowing down the flow of water and allowing it to be gradually released, reducing the strain on drainage systems and nearby water bodies.

MAINTENANCE OF STORM WATER DRAINS

- Effective maintenance of stormwater drains is crucial for ensuring their optimal functionality and mitigating the risks associated with urban flooding.
- By prioritizing the regular maintenance of stormwater drains, communities can mitigate the risks of urban flooding, protect water quality, and ensure the proper functioning of drainage systems. Collaboration, proactive measures, and consistent monitoring are essential for maintaining the integrity and effectiveness of stormwater drains.

